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GEOGRAPHY - 30

Industries in India

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Industries in India

Introduction:

The term **Industry** originates from the Latin word **Industria**, meaning "**diligence**" or "**hard work**." While this definition reflects its general meaning, there is no universal criterion to define all industries. The concept of industry is inherently **ambiguous** because it encompasses **multiple dimensions**. As a result, diverse industries often rely on different criteria for definition.

The most significant elements used to define industries are:

- **Products** (goods and services)
- **Technology and Knowledge**
- **Economic function or position in the value chain**
- **Market**

These four elements frequently overlap and are **not mutually exclusive**. Many industries fit into several of these categories, rather than just one.

Historical Perspective of Indian Industries:

- The **history of industry in India** is as old as human civilization itself. Indian **handicrafts**, crafted in village huts and homes across the country, were highly prized in foreign markets for their quality and aesthetic appeal. These products were created using locally available raw materials, with artisans relying on the **skills and tools passed down through generations**. The result was a long and rich tradition of Indian **artistic handicrafts** of various kinds, known for their intricate workmanship.
- Among the many industries that flourished in early India, the **textile industry**, particularly **cotton textiles**, stood out. This sector enjoyed a place of pride, both within India and in global markets. Historical evidence suggests that Indians were familiar with **weaving as early as 1,500 years before Christ**, a time when Europeans were still using animal skins as clothing. The 17th-century Portuguese writer **Pyrard** recorded that people from the **Cape of Good Hope to China** were dressed in garments made in India. **The fine Dhaka muslin**, a highly sought-after fabric, remained the envy of the world for centuries.
- **India's iron and steel industry** was also highly advanced during this period. A striking example is the **iron column near Qutab Minar in Delhi**, which has stood in the open for over **1,500 years**, exposed to the elements, yet remains **unaffected by corrosion**. This column is a testament to India's **forging and fabricating ingenuity** and symbolizes the technological prowess of ancient Indian metallurgy. It is even believed that the renowned **Damascus swords** were crafted from steel imported from India.
- Apart from cotton textiles and steel, India was famous for various other industries including **wood, stone, and ivory carvings, silk textiles, pottery, bronze, brass, silver, copper works, and dyeing and calico printing**. These industries collectively contributed to India's reputation as a hub of artistic excellence and craftsmanship, which was recognized worldwide.
- **The Industrial Revolution in Europe** introduced **modern factories**, dramatically increasing the scale of manufacturing through **mechanization**. This revolution led to the **migration of workers from villages to cities** as large-scale industries replaced traditional methods of production. The **barter system**, where goods were exchanged for other goods, gave way to a **monetary economy**, with the exchange of goods for money becoming the norm.

- While this revolution brought significant changes in the manufacturing world, it also led to the **decline of traditional village handicrafts and cottage industries** in India. Artisans who once thrived on their craft were unable to compete with the mass-produced, cheaper, and more refined products of modern industries. This resulted in thousands of artisans, such as **goldsmiths, blacksmiths, and weavers**, losing their livelihoods. The once-thriving village industries began to collapse, leading to **widespread unemployment and chaos** in rural areas. Traditional industries, which had formed the backbone of Indian society for centuries, were **eroded by the rise of modern industrialization**.

Factors Influencing the Location of Industries:

The location of industries is influenced by a combination of **geographical** and **non-geographical factors**. While geographical factors like the availability of raw materials, power resources, water, labour, markets, and transport facilities are fundamental, modern industrial development is increasingly shaped by **historical, human, political, and economic factors**. Therefore, factors influencing industrial location can be categorized into **two broad types**:

- **Geographical Factors**
- **Non-Geographical Factors**

Geographical Factors:

Raw Materials:

- The availability of raw materials is a primary determinant of industrial location. Many industries are located near raw material sources, especially when they involve heavy or bulky materials that lose weight during manufacturing or are costly to transport. For example, **jute mills in West Bengal, sugar mills in Uttar Pradesh, and cotton textile mills in Maharashtra and Gujarat** are located close to raw material supplies. Industries like **iron and steel**, which depend on large quantities of coal and iron ore, tend to be situated near the source of these materials.
- On the other hand, industries like **electronics and watch manufacturing**, which use a wide range of light raw materials, are often referred to as "**footloose industries**" because they are not restricted to specific locations based on material availability. These industries can operate in a variety of locations as long as there is sufficient population density and supporting infrastructure.

Power:

- A regular and reliable supply of power is essential for industrial operations. **Coal, mineral oil, and hydro-electricity** are the primary conventional power sources for industries. Industries that rely on large quantities of power, such as **iron and steel or electro-metallurgical and electro-chemical industries**, tend to concentrate near power sources. For instance, **aluminium industries** are typically located near regions producing hydro-electric power, like **Korba and Renukoot**.
- While coal and hydro-power remain critical, the development of long-distance electricity transmission has allowed industries to be located further from power sources. Examples include industries in southern states that moved there following the development of hydro-power in areas that lacked coal resources.

Labour:

- A readily available labour force is essential for industrial development. Industries typically require both **large numbers of workers** and **skilled or technical personnel**. The cost of labour varies depending on the type of industry; for example, labour costs account for about **62%** in clothing-related industries but only **29%** in the chemical industry.

- In India, even with increasing mechanization, many industries still require a large labour force, particularly in **light consumer goods** and **agro-based industries**. These industries tend to be located near large urban centers where unskilled labour is abundant.

Transport:

- Transportation is essential for both the **assembly of raw materials** and the **distribution of finished products**. Historically, the development of railways connecting port towns with their hinterlands significantly influenced the location of industries. For example, the growth of industries around **Kolkata, Mumbai, and Chennai** was driven by the availability of transport facilities. Although transport improvements and industrial development influence each other, the availability of original transport facilities often plays a critical role in determining industrial location.

Market:

- Proximity to markets is crucial for the **distribution of finished goods**. Being close to markets reduces transportation costs, ensures quicker delivery to consumers, and often leads to cheaper goods. This is especially important for **perishable** or **heavy commodities**, where industries tend to be **market-oriented**. The growing importance of market proximity means that many industries are now located near their consumer base unless there is a compelling reason to be located elsewhere.

Water:

- Many industries require large quantities of water, making proximity to **rivers, canals, and lakes** an important consideration. Industries like **iron and steel, textiles, and chemical manufacturing** rely heavily on water for their operations. For instance, **36,400 liters of water** are required to produce just **one kilowatt-hour** of thermal electricity. Additionally, industrial water use often leads to pollution, rendering the water unsuitable for other purposes.

Site:

- The availability of a suitable site is another crucial factor in industrial location. Industrial sites should ideally be **flat**, have **adequate transportation links**, and provide **sufficient space** for factories. In recent times, rising urban land costs have led to a trend of industries being established in rural areas where land is more affordable.

Climate:

- Climate plays a significant role in industrial development. **Extremely hot, cold, humid, or dry climates** are not conducive to industrial growth. For instance, the harsh climate of **northwest India** hinders industrial development, while the **moderate climate of the west coast** fosters industrial growth. About **24% of India's modern industries** and **30% of industrial labour** are concentrated in the **Maharashtra-Gujarat region** due to its favourable climate. Climate is especially important for industries like **cotton textiles**, which require a **humid environment**. In dry climates, **artificial humidifiers** can be used, but they increase production costs.

Non-Geographical Factors:

- In the modern era, **scientific and technological advancements** have led to the use of alternative raw materials, reducing the influence of geographical factors on industrial location. Factors such as the widespread availability of electric power and the increased mobility of labour have further diminished geography's role. Today, **non-geographical factors**, including **economic, political, historical, and social aspects**, play a significant role in influencing the location of industries. The following are some of the key non-geographical factors that determine where industries are set up:

Capital:

- Modern industries are **capital-intensive**, requiring substantial investments. Capital tends to concentrate in **urban centers**, where banking and financial institutions are readily available. Major cities like **Mumbai, Kolkata, Delhi, and Chennai** are thriving industrial hubs because they provide easy access to the capital required for industrial growth. In these cities, financial institutions facilitate the exchange of large sums of money, enabling industrial operations on a large scale.

Government Policies:

- **Government policies** have a profound influence on the **location and distribution of industries**. Through planned interventions, governments aim to **reduce regional disparities**, manage **pollution control**, and prevent the **over-clustering** of industries in metropolitan areas. As part of these policies, industries are increasingly being set up in areas where they can benefit from shared resources such as **water, power**, and mutual supply chains.
- An example of this is the establishment of **industrial estates** across India, catering even to the small-scale industrial sector. **India's Five-Year Plans** have played a **pivotal role in shaping industrial location, promoting the emergence of industries in southern India and backward areas** with growth potential. The establishment of public sector plants has also spurred industrial development in these regions.
- Government initiatives have significantly contributed to the establishment of industries such as **fertilizer factories, iron and steel plants, engineering works, machine tool factories, oil refineries, and defence installations**. Examples of government-driven industrial locations include the **Mathura Oil Refinery, Kapurthala Coach Factory, and Jagdishpur Fertilizer Plant**.

Industrial Inertia:

- **Industrial inertia** refers to the tendency of industries to remain in their original locations, even if the factors that initially influenced their establishment are no longer present. This phenomenon is sometimes called **geographical inertia** or **industrial inertia**. A notable example of this is the **lock industry in Aligarh**, which continues to operate in its original location despite changes in the industry's environment or requirements.

Efficient Organization and Management:

- The success of modern industries depends on **efficient organization and management**. Poor management can result in the mismanagement of capital, leading to financial crises that jeopardize the industry's future. **Effective management** is also essential for handling the labour force. Mismanagement can lead to **labour unrest**, strikes, and lock-outs, all of which negatively affect industrial productivity and may lead to the closure of industries. Therefore, the presence of a competent management team is critical for the smooth functioning and success of any industrial venture.

Banking Facilities:

- Industries require daily transactions involving **large sums of money**, often amounting to crores of rupees. These transactions are only possible in areas with well-developed **banking facilities**. As a result, regions with robust banking systems are more favourable for the establishment of industries.

Insurance:

- Industries face constant risks related to **damage to machinery, materials, and human resources**. To safeguard against these risks, access to **insurance services** is essential. Insurance facilities provide security to industrial enterprises, ensuring that they can recover from unforeseen damages or accidents.

Classification of Industries:

Industries can be **categorized into several types** based on different criteria. These classifications help in understanding the **nature, scale, and structure of industries**, as well as the source of their raw materials and the nature of ownership. Below is a detailed classification of industries.

On the Basis of Strength of Labour:

- **Large Scale Industries** - These are industries that **employ a large number of labourers** within each unit. Industries such as **cotton and jute textile** fall under this category. These industries often involve extensive operations and large production capacities.
- **Medium Scale Industries** - Medium scale industries **neither employ a very large workforce nor a very small one**. Industries like **cycle manufacturing, radio production, and television manufacturing** are examples of medium-scale industries.
- **Small Scale Industries** - Owned and operated by individuals, small scale industries employ only a **small number of labourers**. These industries are typically smaller in terms of production capacity and infrastructure.

On the Basis of Raw Materials and Finished Goods:

- **Heavy Industries** - These industries utilize heavy and **bulky raw materials** and produce goods of a similar nature. A prime example is the **iron and steel industry**, which processes large quantities of raw material to produce heavy finished products.
- **Light Industries** - Light industries, in contrast, use **lighter raw materials** to produce lighter goods. Examples include the manufacture of **electric fans and sewing machines**.

On the Basis of Ownership:

- **Private Sector Industries** - These industries are **owned by individuals or private firms**. Examples include **Bajaj Auto** and **TISCO** (Tata Iron and Steel Company), which is based in Jamshedpur.
- **Public Sector Industries** - These are industries **owned by the state or government agencies**. Well-known examples include **Bharat Heavy Electricals Ltd. (BHEL)**, **Bhilai Steel Plant**, and **Durgapur Steel Plant**.
- **Joint Sector Industries** - In these industries, **ownership is shared between private firms and the government** or its agencies. Examples are **Gujarat Alkalies Ltd.** and **Oil India Ltd.**
- **Co-operative Sector Industries** - These industries are **owned and operated cooperatively by a group of people**, usually those who supply raw materials for the industry. An example is a **sugar mill** that is owned and run by farmers.

On the Basis of Source of Raw Materials:

- **Agro-Based Industries** - These industries rely on **agriculture for their raw materials**. Examples include the **cotton textile industry, jute textile industry, sugar industry, and vegetable oil industry**.
- **Mineral-Based Industries** - These industries use **minerals as their primary raw materials**. The **iron and steel, aluminium, and cement** industries fall into this category.
- **Pastoral-Based Industries** - These industries obtain their **raw materials from animals**. Examples include industries related to **hides, skins, bones, horns, dairy products, and shoe-making**.

- **Forest-Based Industries** - Industries that use **raw materials from forests**, such as **paper, cardboard, lac, rayon, resin, and tanning of leather**, are classified under this category.

Miscellaneous Industries:

- **Footloose Industries** - **Footloose industries** refer to those industries that **do not have a significant locational preference**, as their **input resources and output markets are readily available** in various locations. This flexibility in location allows them to establish operations in different places, hence the term "**footloose**." Unlike other industries tied to specific resource or market locations, footloose industries **enjoy greater mobility** due to the availability of essential inputs across a wide geographical area.
 - One of the key characteristics of footloose industries is that they generally require **smaller plant sizes** compared to both heavy and small-scale industries. This compact nature makes them adaptable to a variety of locations. Some well-known examples include **diamond cutting, precision electronics, watchmaking, computer chip manufacturing, and mobile phone production**. These industries often produce **low-volume, high-value outputs**, which further supports their flexibility in location. Additionally, **footloose industries are generally non-polluting**, contributing to their ability to operate in diverse environments without environmental constraints.
- **Sunrise Industries** - **Sunrise industries** are industries that exhibit the potential for **substantial and rapid growth**. These industries are often in the **early stages of development** and are characterized by a **high rate of innovation, entrepreneurship, and significant investments**, particularly from start-ups and venture capitalists. The term "**sunrise**" symbolizes the **new and emerging nature** of these industries, as they are relatively new or have only recently gained prominence in the market.
 - The key feature of a sunrise industry is its **fast-paced growth**. This growth is driven by technological advancements and changing consumer demands. Examples of sunrise industries include the **food processing industry, petrochemical industry, hydrogen fuel production, space tourism, and online encyclopaedias**. These industries represent sectors of the economy that are expected to play a crucial role in future economic development due to their innovative nature and capacity for rapid expansion.
- **Weight Losing Industries** - **Weight losing industries** are those where the **raw materials are bulky but the finished products are relatively light**. Due to the weight loss during the production process, these industries are typically located near the source of raw materials to reduce transportation costs for bulky inputs. The **proximity to raw materials is essential for weight losing industries to optimize efficiency and minimize expenses** related to moving heavy raw materials over long distances.
 - A classic example of a weight losing industry is the **sugar industry**, which is typically situated near sugarcane-producing regions. This is because **sugarcane is bulky**, and the process of extracting sugar results in a much lighter final product. Other examples include the **pulp industry, copper smelting, and pig iron industries**, all of which require raw materials that are significantly heavier than their finished goods. These industries rely on being close to their resource base to reduce production and transportation costs.
- **Village Industries** - These industries are **located in rural areas** and cater primarily to the needs of the rural population. Examples include industries involved in **oil extraction, grain grinding**, and the production of **agricultural implements**.

- **Cottage Industries** - Set up by **artisans in their own homes**, these industries involve working with materials like **wood, cane, brass, and stone**. Examples include **handloom, khadi, and leatherwork**.
- **Consumer Goods Industries** - These industries **convert raw materials or primary products** into goods **directly used by consumers**. Examples include the **textile industry, bakery products, and the sugar industry**.
- **Ancillary Industries** - These industries **manufacture parts and components that are used by larger industries to produce heavy articles**, such as **trucks, buses, railway engines, and tractors**.
- **Basic Industries** - Industries upon which **other industries depend for their manufacturing processes** are called basic industries. The **iron and steel industry** and **power generation industry** are key examples.
- **Capital-Intensive Industries** - Industries that **require large capital investments** are classified as capital-intensive. Examples include the **iron and steel, cement, and aluminium** industries.
- **Labour-Intensive Industries** - These industries **rely heavily on labour** for their operations. In such industries, the workforce is more significant than capital. Examples include **shoe-making and bidi-manufacturing**.

Industrial Regions:

Industrial regions are **geographical areas where industries concentrate** due to **favorable socioeconomic conditions**. These regions are characterized by **large-scale manufacturing activities**, which significantly contribute to local employment and economic development. They typically develop around **natural resources** such as coal, iron ore, or major water supplies, as these resources provide essential raw materials and energy for industrial processes. Furthermore, **well-established transportation networks**, especially **railways**, play a crucial role in supporting the movement of raw materials and finished goods, facilitating industrial growth.

Industrial regions exhibit a **diversity of industries**, with various unrelated production types coexisting. These regions are not limited to a single industry but may include sectors such as **steel manufacturing, electricity generation, primary food processing**, and even **service sectors like education**. This industrial variety arises from the need for a large and diverse **labor force**, which drives the expansion of residential areas and **mixed land use** near industrial hubs.

The formation of industrial regions is driven by the **clustering of industries**. This phenomenon occurs when **industries group together to benefit from proximity**, which offers advantages like **reduced transportation costs, access to a skilled workforce, and the sharing of infrastructure**. Geographers emphasize that **specific geographic characteristics**, such as **access to resources, transportation, and energy**, play a key role in determining the locations of these industrial clusters.

Identifying Industrial Clusters:

Geographers use several **indices** to detect the clustering of industries within a region. **These indices include:**

- The **number of industrial units** operating in the area.
- The **total number of industrial employees**, reflecting the workforce engaged in manufacturing and related activities.
- The **population engaged in secondary activities**, which refers to those involved in manufacturing, construction, and processing industries.
- The **proportion of industrial employees to the total workforce**, which helps assess the relative significance of industrial employment in the region.
- The **amount of power utilized for industrial purposes**, as higher power consumption often correlates with more extensive industrial activities.

In addition to these indices, geographers assess the **total industrial production** within a region, typically measured through the **gross industrial output**. This metric captures the value-added by manufacturing activities, serving as a key indicator of the economic importance of an industrial region.

Major Industrial Regions of India:

- Mumbai-Pune Industrial Region
- Bangalore-Tamil Nadu Industrial Region
- Hugli Industrial Region
- Ahmedabad-Vadodara industrial region
- Chotanagpur Industrial Region
- Vishakhapatnam-Guntur Industrial Region
- Gurgaon-Delhi-Meerut Industrial Region
- Kollam-Thiruvananthapuram Industrial Region



Mumbai-Pune Industrial Region:

- The **Mumbai-Pune Industrial Region** is recognized as **India's foremost industrial hub**, with its development deeply rooted in the **British colonial influence**. The foundation of this region's industrial prominence began with the **establishment of the Mumbai harbor**, which provided crucial port facilities for trade and industry.
- A significant aspect of the region's industrial growth is its close association with the history of **India's cotton textile industry**, which became a major force in driving industrialization. Several factors have contributed to this growth, including the region's **humid environment**, which is ideal for textile production, the **natural port facilities** at Mumbai, the availability of **abundant hydro-power** from nearby rivers, **skilled labor**, and a vast **hinterland for cotton production**.
- Greater **Mumbai** alone is home to approximately **8,000 registered factories**, with **350 dedicated to cotton textile production**. However, the region is not limited to textiles alone; it boasts a diverse industrial landscape, including **engineering products, chemical industries, food processing, leather goods, pharmaceuticals**, and even the **film industry**. The primary industrial output in Mumbai consists of **light-textured, fine, and ultra-fine cotton fabrics**, employing around **15 lakh people**.
- In the wider region, **Pune** serves as the second most significant industrial center, hosting around **1,200 registered industries**. Pune's manufacturing base includes a range of industries producing **metallurgical, chemical, engineering, and automotive products**. Notably, Pune houses two major plants specializing in the **production of scooters and mopeds**.
- **Challenges Facing the Mumbai-Pune Industrial Region:**
 - **Inadequate power supply**
 - **Outdated machinery** in many industries
 - **High land costs** and **expensive commercial spaces**
 - Increasing **workplace unrest** and **growing regionalism**
 - A **high crime rate** in some areas
 - Rising levels of **environmental pollution**

- Additionally, the **partition of India in 1947** negatively impacted the region, as **Pakistan received a significant portion of the irrigated cotton acreage**, affecting the production of **long-staple cotton**. Moreover, **Mumbai**, the central hub of the region, is constrained by a **lack of space for further industrial expansion**. To address these challenges and alleviate congestion, the **dispersal of industries** into less saturated areas is crucial for ensuring **sustainable growth**.

Bangalore-Tamil Nadu Industrial Region:

- The **Bangalore-Tamil Nadu Industrial Region** spans across **Tamil Nadu** and the southern parts of **Karnataka**, experiencing remarkable growth, particularly **post-independence**. This region is primarily recognized for its **cotton production**, which has been a driving force behind its industrial development.
- Several factors have contributed to the prosperity of this region, including a **disciplined labor force** (both skilled and unskilled), a reliable **electricity supply** from sources such as **Mettur, Papanasam, Pykara, Savitri**, and **Sivasamudram**, and the region's **strategic proximity to seaports** like **Chennai, Kochi, Mangalore**, and **Tuticorin**.
- The **textile industry** dominates this region's industrial landscape, employing around **60% of the workforce**. Other significant industries include **engineering** (18%) and **food processing** (12%). The growth of the **cotton mills** has also greatly benefited the **loom business**. Furthermore, **Bangalore** has emerged as a significant center for **heavy engineering enterprises**.
- Prominent industrial establishments in the region include **HAL (Hindustan Aeronautics Limited)**, **machine tools**, **HTL (Hindustan Teleprinters Limited)** telephones, and **Bharat Electronics**. The region also produces a wide variety of goods, including **textiles, rail wagons, diesel engines, radios, light engineering products, rubber goods, pharmaceuticals, aluminum, sugar, cement, glass, paper, chemicals, film, cigarettes, matchboxes, and leather goods**.
- Recent industrial developments in the region include the establishment of a **petroleum refinery near Chennai**, an **iron and steel mill in Salem**, and several **fertilizer factories**. The region continues to evolve as a **dynamic industrial powerhouse**, contributing significantly to the economic landscape of **Tamil Nadu** and **southern Karnataka**.

Hugli Industrial Region:

- The **Hugli industrial zone**, located along the banks of the **Hugli River**, has seen significant industrial development, especially in the western region of **Midnapur**. This growth can be attributed to several favorable factors. The **strategic location along the Hugli River** has provided access to easy transportation and export facilities. Additionally, the region is well-supplied with **agro-raw materials** such as **jute, indigo, and tea**, and is in close proximity to the **coal mines of Raniganj and Jharia**, which fuel many of its industries. The availability of **abundant water, cost-effective labor**, and the region's historical significance as the capital of British India (from **1773 to 1911**) also contributed to the early establishment of industries in Kolkata.
- This industrial belt specializes in the production of a wide variety of goods, including **jute, silk, and cotton textiles**, as well as **engineering products, electrical goods, automobiles, chemicals, pharmaceuticals, transport equipment, leather footwear, iron and steel, food processing, light machinery, locomotives, and machine spare parts**. Major industrial cities and towns within this region include **Naihati, Bhatpara, Shamnagar, Krishnanagar, Serampore, Titagarh, Rishra, Kolkata, Haora, and Budge**.

- Despite its industrial success, the Kolkata-Hugli region faces several challenges. These include **traffic bottlenecks**, **limited space for expansion**, **water scarcity**, **sanitation and infrastructure deficiencies**, and the **silting of the Hugli River**, which affects operations at **Kolkata port**. Additionally, the region struggles with **outdated machinery**, **political turmoil**, labor issues such as **lockouts and strikes**, the **Naxalite movement**, and frequent **power supply failures**. In response to these challenges, the **West Bengal government** has been actively pursuing **liberalization policies** and encouraging both **domestic and global investments** to address these issues and revitalize the region.

Ahmedabad-Baroda Industrial Region:

- The **Ahmedabad-Baroda region** is the third-largest industrial zone in India. This area benefits from various factors that have fueled its industrial success, including the **availability of cotton** in the surrounding hinterland, **affordable land**, and a pool of **skilled and unskilled labor**. Additionally, **port facilities** and the proximity to multiple **power stations** have contributed to its rapid industrialization. This region is particularly known for its **cotton textile industry** and has also made significant strides in the **chemical, technical, and medicinal sectors**.
- There are around **11,000 registered industries** in this region, employing over **15 lakh people**. Key industrial centers include **Vadodara**, known for **woolen textiles** and **petrochemical products**, and **Surat**, which is famous for its **silk textiles** and **diamond cutting**. Other prominent cities contributing to the industrial landscape of the region include **Anand, Ankleshwar, Bhavnagar, Bharuch, Godhra, Jamnagar, Kalol, Kheda, Rajkot, and Surendranagar**.
- However, the region faces challenges such as **water scarcity** and a **shortage of high-quality cotton**, which have posed difficulties for the textile industry. Additionally, **communal tensions** in recent years have also created obstacles for new industrial investments.

Chhotanagpur Industrial Region:

- The **Chhotanagpur region**, spanning parts of **Jharkhand, Odisha, southern Bihar, and western West Bengal**, is often referred to as the "**Ruhr of India**" due to its significant concentration of **iron and steel industries**. This region is rich in **fossil fuels** and **metallic and nonmetallic materials**, making it an industrial powerhouse. The presence of the **Damodar Valley Corporation** ensures a steady supply of **electricity**, which powers much of the region's industrial activities.
- This industrial zone is home to major hubs for **iron and steel production**, including **Asansol, Bokaro, Burnpur, Durgapur, Kulti, Jamshedpur, and Rourkela**. In addition to steel, the region is known for industries such as **fertilizer production** in **Sindri**, **locomotive manufacturing** in **Chittranjan**, **HMT machinery** in **Ranchi**, and **glass production** in **Ramgarh and Bhurkunda**. The **Damodar Valley** is also notable for its thermal and hydroelectric projects, which help meet the region's electricity demands.
- The region's proximity to densely populated areas provide access to **affordable labor**, and the **Hugli region** serves as a significant market for the products manufactured here. However, despite its industrial potential, the region struggles with **power shortages**, **political unrest**, and the influence of **Naxalite movements**. Frequent **labor upheavals** have deterred some investors, adding complexity to industrial growth in this resource-rich region.

Visakhapatnam-Guntur Industrial Region:

- The industrial belt extending from **Visakhapatnam** to **Kurnool** and **Prakasam** in southern India has grown significantly due to several strategic factors. This region's development is anchored by its proximity to the **ports of Visakhapatnam** and **Machilipatnam**, which provide vital links for trade and industrial activities. Additionally, the region benefits from the **fertile agriculture** and **mineral resources** available in the hinterlands, which support the growth of various industries.
- A key advantage of this region is the availability of energy from the **Godavari basin coalfields**, which ensures a reliable energy supply for its industries. The **shipbuilding industry** in Visakhapatnam, established in **1941**, plays a significant role in the region's industrial heritage. Furthermore, the establishment of **petroleum refineries**, which primarily process **imported petroleum**, has driven the expansion of the **petrochemical industry**, further catalyzing industrial growth.
- The region is home to a wide range of industries, including **sugar, textiles, jute, paper, fertilizer, cement, aluminum, and light engineering**. The key industrial centers in this zone are **Visakhapatnam, Vijayawada, Vijaynagar, Rajahmundry, Guntur, Eluru, and Kurnool**. This region, with its **strategic location** and **natural resource base**, has emerged as a vital hub for various industrial activities, contributing significantly to the region's economic development.

Gurgaon-Delhi-Meerut Industrial Region:

- Although located far from major **mineral and energy resources**, the **Gurgaon-Delhi-Meerut** region has carved out a unique industrial identity by catering to **local market demands**. This industrial zone is dominated by relatively small-scale industries, primarily involved in **electronics, light engineering, and electrical goods**.
- Additionally, the region boasts significant industries in **textiles**, including **cotton, woolen, synthetic textiles**, and **hosiery**, as well as other sectors such as **sugar, cement, machine tools, tractors, motorcycles, agricultural equipment, chemicals, and vanaspati**. More recently, the **software industry** has emerged as a key player in the region, further diversifying the industrial landscape.
- To the south, the **Agra-Mathura industrial sector** specializes in the production of **glass and leather goods**, with **Mathura** housing a major **petrochemical complex** thanks to its oil refinery. Important industrial cities contributing to the region's economy include **Gurgaon, Delhi, Shahdara, Faridabad, Meerut, Modinagar, Ghaziabad, Ambala, Agra, and Mathura**.
- Despite its industrial success, the region faces several challenges, such as the **high cost of land, transportation congestion, and rising crime rates**. However, the diversity of industries and the focus on local market needs continue to drive the region's economic growth.

Kollam-Thiruvananthapuram Industrial Region:

- The industrial region of **Kollam-Thiruvananthapuram** spans the districts of **Thiruvananthapuram, Kollam, Alwaye, Ernakulam, and Alappuzha**. This area is distinct from the mineral-rich belts of India, as its industrial development is grounded in **plantation agriculture** and **hydropower resources**. Industries in this region primarily focus on **agricultural goods processing** and **market-oriented light industries**.
- The key industries in this region include **cotton textiles, sugar production, rubber processing**, the manufacture of **matchboxes, glass production, chemical fertilizers**, and industries related to **fish-based products**. Additionally, the

region is notable for its **food processing** industries, as well as the production of **paper, coconut coir products, aluminum, and cement**.

- The main industrial centers in this region are **Kollam, Thiruvananthapuram, Aluva, Kochi, Alappuzha, and Punalur**. The industries in these centers play a crucial role in contributing to the region's **economic growth**, driven by the natural resources and strategic industries developed within this area.

Minor Industrial Regions of India:

- India has several **minor industrial regions** that contribute significantly to the country's economy by producing a variety of goods. These regions are spread across different states and are known for their specialized industries. Below is a detailed account of the key industrial regions, their main products, and the major industrial centers within them.

Kanpur-Lucknow Industrial Region:

- This region is one of the prominent **industrial zones in northern India**, with Kanpur and Lucknow as the major industrial centers. It is renowned for its production of **cotton, woolen, and jute textiles**, making it a significant hub for textile manufacturing. Apart from textiles, the region also produces **leather goods, fertilizers, chemicals, pharmaceuticals, and light machinery**. **Electric goods** are also manufactured extensively here, contributing to the region's diverse industrial output.

Assam Valley Industrial Zone:

- The **Assam Valley** is known for its diverse range of industries, including **petrochemical, jute and silk textiles, and tea processing**. Additionally, industries in this region also focus on **paper, plywood, match production, and food processing**. Major cities driving industrial growth in this zone include **Bongaigaon, Dibrugarh, Digboi, Guwahati, Noonmati, and Tipsukia**. Assam's vast natural resources, including oil, have enabled the growth of the petrochemical industry, while its tea industry is globally recognized.

Darjeeling-Siliguri Industrial Region:

- The **Darjeeling-Siliguri region** is primarily known for its **tea processing industry**, which has earned global acclaim for producing some of the finest teas. The region also thrives on **tourism**, which significantly boosts the local economy, especially in the scenic hill areas of Darjeeling.

North Bihar and Eastern Uttar Pradesh Industrial Region:

- This industrial region covers parts of **North Bihar and Eastern Uttar Pradesh**, where industries are primarily focused on **sugar production**, owing to the region's rich sugarcane cultivation. Other major industries include **cement, glass, jute, fertilizers, and locomotive production**. The region is also engaged in **paper and food processing**. Key industrial hubs include **Allahabad, Dalmianagar (Bihar), Gorakhpur, Patna, Sultanpur, and Varanasi**.

Indore-Ujjain Industrial Region:

- The **Indore-Ujjain region** in Madhya Pradesh is a growing industrial hub, specializing in **cotton textiles, chemicals, medicines, and electrical and technical items**. The region also has a significant **food processing industry**, making it an important industrial zone in central India.

Amritsar-Jalandhar-Ludhiana Industrial Region:

- This region in **Punjab** is famous for its **sports products**, particularly in **Jalandhar**, which is one of the leading centers for sports goods production in India. The region also excels in the production of **cotton and woolen textiles, hosiery**, and **food processing**. In addition, **tourism** is a growing sector in the region, with **Amritsar** being a major religious and cultural destination.

Nagpur-Wardha Industrial Region:

- The **Nagpur-Wardha region** in Maharashtra is another key industrial area, with industries focusing on **textiles, engineering, chemicals, and food processing**. Nagpur, being centrally located, has emerged as an important hub for industries serving central and western India.

Godavari-Krishna Delta Industrial Region:

- The **Godavari-Krishna Delta** in Andhra Pradesh is rich in agricultural and industrial activities. The region's dominant industries include **iron and steel production, shipbuilding, and fertilizers**. Other key industries are **rice milling, cotton textiles, sugar, fish processing, engineering, and chemicals**. Important industrial centers include **Guntur, Machilipatnam, Rajahmundry, and Visakhapatnam**.

Dharwad-Belgaum Industrial Region:

- This region in Karnataka has a well-established industrial base, specializing in **cotton textiles, chemicals, and spice packing**. It is also a hub for **food processing**, capitalizing on the agricultural output of the surrounding areas.

Kerala Coast Industrial Region:

- The **Kerala Coast** is home to several industries, with **Kochi** as the key industrial center. **Coconut oil extraction, rice milling, and fish packaging** are the primary industries here. The region is also known for the production of **paper, coir matting, and shipbuilding** at Kochi, along with **petroleum refining**. Kochi's role in **chemical and electronic goods production** further enhances the industrial profile of the Kerala coast.

Major Industries in India:

Iron and Steel Industry:

- The **iron and steel industry** plays a pivotal role in industrial development, as it is considered a **basic industry** essential for the growth of other industries. It is classified as a **weight-losing industry**, where the weight of the finished product is less than the weight of the raw materials. Hence, iron and steel plants are often located near raw material sources to minimize **transportation costs**. The primary raw materials required for this industry include **iron ore and coking coal**, with additional materials such as **limestone, dolomite, and manganese**.
- **The first significant effort to establish the iron and steel industry** in India was in **1874** when the **Bengal Iron Works** was set up in **Kulti**, West Bengal. Later, in **1880-81**, the **Public Works Department** of the Government of India took over the construction and renamed it **Barakar Iron Works**. In **1907**, the **Tata Iron and Steel Company (TISCO)** was founded by **Jamshedji Tata** in **Jamshedpur (Sakchi)**. It was a major milestone in the private sector, with pig iron production beginning in **1911** and steel production starting in **1913**.
- Other early developments include the establishment of an iron and steel plant in **Hirapur (Asansol, West Bengal)** in **1908**. In **1918**, the **Indian Iron and Steel Company (IISCO)** was established in **Burnpur, West Bengal**, and by **1936**, the Kulti and Hirapur plants were integrated into IISCO. Another early plant, **Vishweshwarayya Iron and Steel Limited (VISL)**, was set up in **Bhadravati, Karnataka** in **1923**.

Iron and Steel Industry After Independence:

- After India's independence, **significant growth in the iron and steel industry** occurred during the **Second Five-Year Plan**, when three major plants were established:
 - **Bhilai (Chhattisgarh)** - Built with assistance from **Russia**.
 - **Durgapur (West Bengal)** - Set up with help from **Britain**.
 - **Rourkela (Odisha)** - Established with cooperation from **Germany**.
- In the **Third Five-Year Plan**, it was decided to set up the **Bokaro Steel Plant** in **Bokaro, Jharkhand**, which began production in **1974** with assistance from **Russia**.
- In **1973**, the **Steel Authority of India Limited (SAIL)** was established as a public sector undertaking to oversee the development of the iron and steel industry. Presently, plants in **Burnpur (IISCO)**, **Durgapur**, **Rourkela**, **Bhilai**, **Bokaro**, **Bhadravati (VISL)**, and **Salem** come under SAIL.

Based on their geographical location, iron and steel plants in India can be categorized into four groups:

- **Plants in Coalfields:**
 - **Kulti-Burnpur-Hirapur (IISCO)**, **Durgapur**, and **Bokaro** are located in the **Damodar river valley** coalfields.
- **Plants in Iron Ore Areas:**
 - **Bhilai (Chhattisgarh)**, **Rourkela (Odisha)**, **Bhadravati (Karnataka)**, **Salem (Tamil Nadu)**, and **Vijayanagar (Karnataka)** are established in areas rich in iron ore deposits.
- **Plants between Coal and Iron Ore Fields (Ideal Location):**
 - **Jamshedpur (TISCO)** is situated in an ideal location between coal and iron ore areas.
- **Plants in Coastal Regions:**
 - **Vishakhapatnam (Andhra Pradesh)** and **Gopalpur (Odisha)** are located in coastal regions to facilitate easy import of raw materials and export of finished products.

Major Iron and Steel Plants in India:

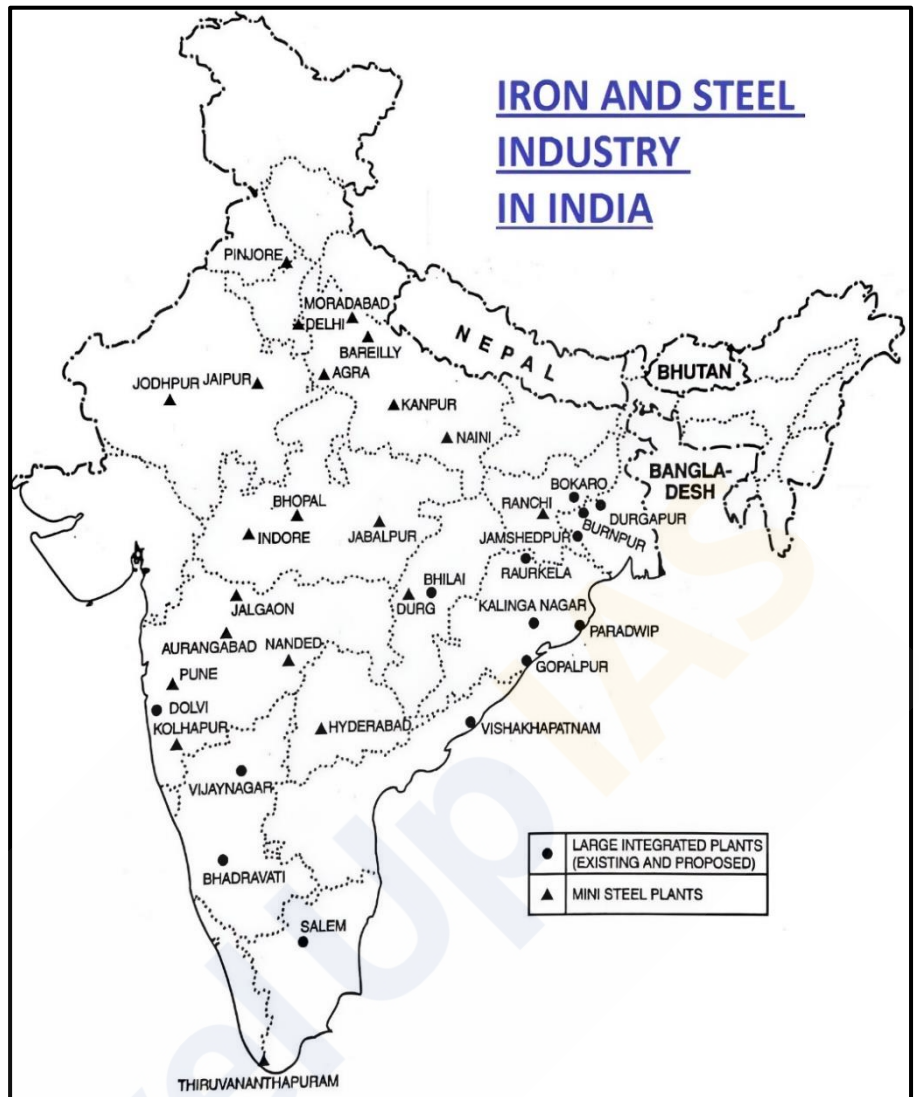
- **Kulti-Burnpur-Hirapur (IISCO):**
 - Located near **Asansol** in West Bengal, these plants receive **iron ore from Singhbhum (Jharkhand)**, **coal from Raniganj, Jharia, and Ramgarh**, **limestone from Birmittapur and Gangpur (Odisha)**, and **manganese from Barajamda-Banspani (Jharkhand-Odisha)**. The plants are well connected to markets via the **Asansol-Kolkata** rail route.
- **Tata Iron and Steel Company (TISCO), Jamshedpur:**
 - TISCO's location between iron ore and coal fields is considered **ideal**. The plant sources **iron ore from Noamundi (Singhbhum)** and **Badam Pahar (Odisha)**, **coking coal from Jharia**, and **limestone and dolomite from Birmittapur (Odisha)**. Fresh water is supplied by the **Dimna dam**, and the plant is well connected to **Kolkata harbor**, which is 240 km away.
- **Vishweshwarayya Iron and Steel Limited (VISL), Bhadravati:**
 - VISL, located in **Shimoga district (Karnataka)**, primarily uses **iron ore from Baba Budan hills**. Initially, **charcoal** was used as fuel until **electric furnaces** were introduced in 1951, powered by electricity from the **Jog hydel project**.

- **Rourkela Steel Plant, Odisha:**

- Set up with German collaboration in **1959**, the Rourkela plant obtains **iron ore from Kendujhar** and **Sundargarh**, **coal from Jharia**, and **manganese from Kendujhar**. It is powered by electricity from the **Hirakud project** and has access to water from the **South Koel and Sankh rivers**. The plant is strategically located on the **Kolkata-Mumbai rail route**.

- **Bhilai Steel Plant, Chhattisgarh:**

- Established with Russian assistance in **1959**, this plant is located in **Durg, Chhattisgarh**. It sources **iron ore from Dalli Rajhara**, **coal from Korba and Kargali**, and **manganese from Balaghat-Bhandara**. The plant is powered by the **Korba Thermal Power Station** and gets water from the **Tendula dam**.



- **Durgapur Steel Plant, West Bengal:**

- Set up with British cooperation in **1962**, the Durgapur plant benefits from the **Raniganj-Jharia coal belt**. It also sources **iron ore from Noamundi (Singbhum)** and power from the **Damodar Valley Corporation**.

- **Bokaro Steel Plant, Jharkhand:**

- Established in **Bokaro (Jharkhand)** with Russian assistance, this plant began operations in **1974**. It sources **coal from Jharia and Bokaro**, **iron ore from Kendujhar**, and **limestone from Palamau**. The plant gets **water and hydroelectricity** from the **Damodar Valley Corporation (DVC)**.

- **Salem Steel Plant, Tamil Nadu:**

- This plant, set up in **1982**, specializes in producing **stainless steel**. It obtains **iron ore from Shevaroy hills**, **lignite from Neyveli coal mines**, and **local manganese and limestone**.

- **Vishakhapatnam Steel Plant, Andhra Pradesh:**

- **Vishakhapatnam Steel Plant**, located in Andhra Pradesh, holds the distinction of being **India's first coastal steel plant**. It was established by the **Rashtriya Ispat Nigam Limited (RINL)** during the **6th Five-Year Plan**. This plant plays a pivotal role in India's steel production, sourcing its key raw materials from various regions. **Iron ore** is obtained from the **Bailadilla mines in Chhattisgarh**, while **coal** is sourced both domestically from the **Damodar Valley** and internationally from **Australia**. In addition, **limestone**, another critical input, is procured from the **Khammam district in Telangana**. The strategic location of this plant near the coast allows for efficient transportation and import of raw materials, which contributes to its operational success.

Steel Plants Set Up in the Private Sector in India:

- **Tata Iron and Steel Company (TISCO)** - Located in **Jamshedpur, Jharkhand**, TISCO is one of India's oldest and most prominent private steel companies.
- **Essar Steel Limited** - This steel plant is located in **Hazira**, in the **Surat district of Gujarat**. It is a major player in the Indian steel industry.
- **Ispat Industries Limited (IIL)** - Situated in **Dolvi**, in the **Raigarh district of Maharashtra**, IIL has since been merged with **JSW Steel Ltd**, a company owned by the **Jindal Group**, further consolidating its position in the steel sector.
- **Jindal Vijaynagar Steel Limited (JVSL)** - This plant is located in **Toranagallu**, within the **Bellary-Hospet belt of Karnataka**. Like IIL, JVSL is also a unit of **JSW Steel Ltd**, part of the Jindal Group.

Aluminium Industry:

- The aluminium industry requires **large-scale availability of electricity**, making it essential for aluminium plants to be located in regions where **cheap electricity** is accessible. India's aluminium industry has evolved through several Five-Year Plans, strategically placing plants near sources of both bauxite and electricity.
- The first aluminium plant in India was established at **Jaykay Nagar in West Bengal**. During the **Second Five-Year Plan**, two more plants were set up at **Hirakud in Odisha** and **Renukoot in Uttar Pradesh**. Later, during the **Third Five-Year Plan**, an aluminium plant was established at **Mettur in Tamil Nadu**, followed by the establishment of a plant at **Belgaum in Karnataka** during the **Fourth Five-Year Plan**.

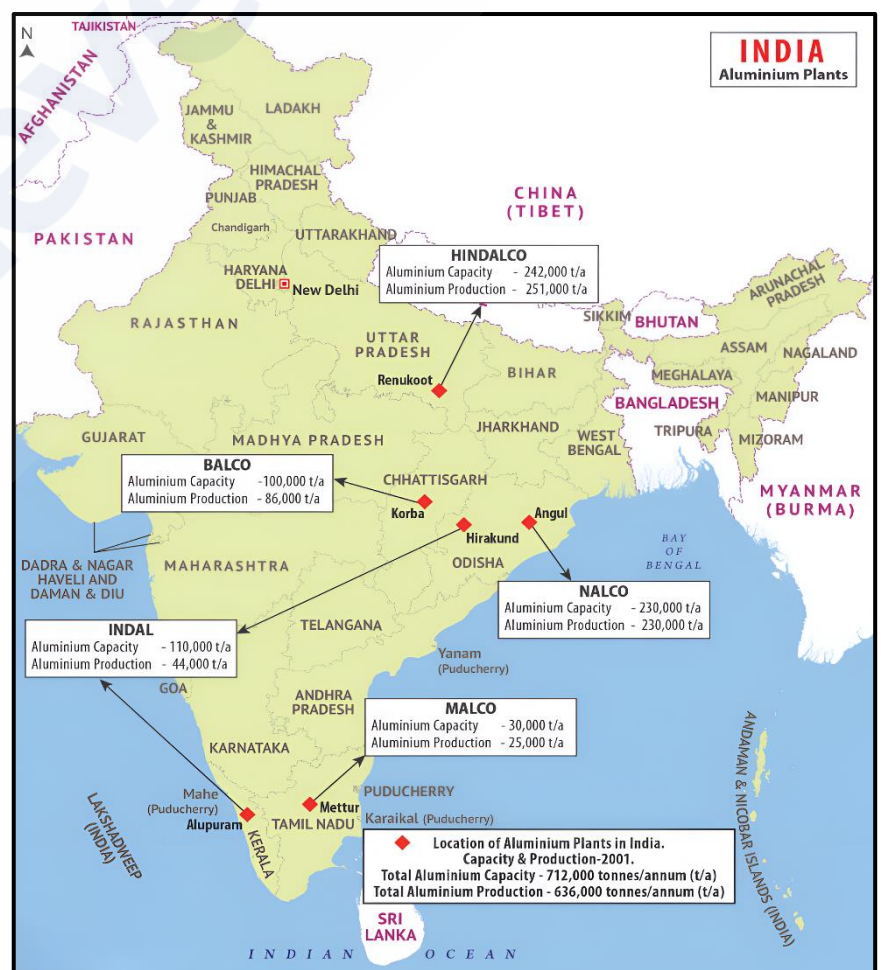
Major Aluminium Companies and Their Plants:

• Hindustan Aluminium Company (HINDALCO):

- The **Renukoot plant in Uttar Pradesh** receives bauxite from the mines of **Ranchi, Palamau, and Lohardaga**. It benefits from **cheap electricity from the Rihand Hydel Power Project**.

• Indian Aluminium Company Limited (INDALCO):

- **Muri (Jharkhand)** - The plant sources bauxite from **Lohardaga** and coal from the **Damodar Valley**, producing alumina from bauxite.
- **Alwaye (Kerala)** - This plant manufactures aluminium from alumina, which is brought from the Muri plant. It utilizes hydro-electricity from the **Pallivasal project**.



- **Belur (West Bengal)** - The plant specializes in producing aluminium sheets from aluminium lumps sourced from Alwaye.
- **Hirakud (Odisha)** - Aluminium is made from alumina obtained from Muri, and the plant draws power from the **Hirakud hydro-electric project**.
- **Belgaum (Karnataka)** - Another important aluminium production facility under INDALCO.
- **Bharat Aluminium Company Limited (BALCO):**
 - **Korba (Chhattisgarh)** - This plant uses bauxite from the **Amarkantak hills** and electricity from a local **NTPC plant**. Post-disinvestment, 51% of BALCO's shares were acquired by **Sesa Sterlite Ltd**, part of the Vedanta group, while 49% remains with the Government of India. Vedanta has also set up a plant in **Jharsuguda (Odisha)**.
 - **Koyna (Maharashtra)** - This plant uses bauxite from Maharashtra and electricity from the **Koyna hydro-electric plant**.
- **Madras Aluminium Company (MALCO):**
 - **Mettur (Tamil Nadu)**: Bauxite is sourced from the **Shevaroy hills**, and hydro-electric power comes from the **Mettur project**.
- **National Aluminium Company Limited (NALCO):**
 - **Damanjodi (Koraput, Odisha)**: This plant uses bauxite from the **Panchpatmali mine** in Koraput.
 - **Angul (Dhenkanal, Odisha)**: The plant draws water from the **Brahmani river** and coal from the **Mahanadi Valley** region.
- **Aluminium Corporation of India:**
 - **Jaykay Nagar (West Bengal)**: Bauxite for this plant comes from **Lohardaga**, while coal is sourced from the **Damodar Valley**.
- **Vedanta Aluminium Ltd. (Jharsuguda, Odisha):**
 - **Vedanta** operates a significant aluminium production facility at **Jharsuguda, Odisha**, making it one of the major players in the Indian aluminium industry.

Cement Industry:

- The **cement industry** is a fundamental sector of the Indian economy, serving as a **backbone for infrastructure development**. The organized production of cement in India began in **1904** in **Madras**, marking the inception of this crucial industry. Cement is classified as a **weight-losing industry**, meaning that it loses weight during the manufacturing process, which necessitates its localization near raw materials to minimize transportation costs.
- The primary **raw materials** used in cement production include **coal**, **limestone**, and **gypsum**. The availability of **sufficient limestone reserves** has led to the industry's concentration in states like **Madhya Pradesh**, where the production has flourished. In recent years, alternative raw materials such as **sea shells**, **sludge** from chemical fertilizer production, and **slag** from the iron and steel industry have also been integrated into the cement manufacturing process.
- **India** stands as the **second-largest producer** of cement globally as of the year **2020**, reflecting the industry's critical role in the country's economic development and infrastructure projects.

Raw Material Sources:

- **Sea Shells** - Used in plants located in **Dwarika, Chennai, Thiruvananthapuram, and Porbandar.**
- **Sludge** - The sludge from the fertilizer plant in **Sindri** serves as a base for the cement plant there.
- **Slag** - Cement plants near iron and steel plants include **Jhinkpani** (Chaibasa, Jharkhand), **Bhadravati, Rourkela, Durgapur, Vishakhapatnam, and Durg** (Chhattisgarh).

Geographic Concentration:

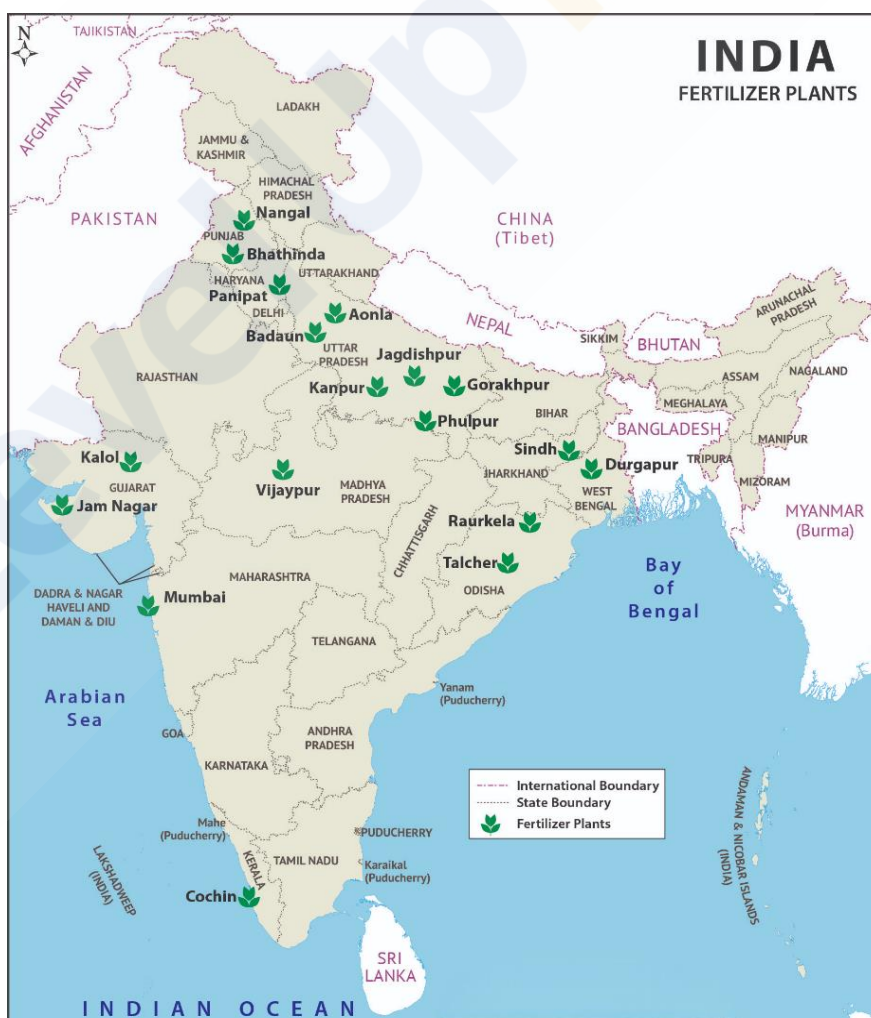
- The centralization of the cement industry is particularly prominent in states with abundant limestone reserves, such as **Madhya Pradesh, Andhra Pradesh, Rajasthan, Tamil Nadu, Bihar, and Karnataka.**
- **Major Cement-Producing States and Their Plants:**
 - **Madhya Pradesh** - Major plants include **Satna, Katni, Maihar, Jabalpur, Ratiam, Neemach, Banmor, and Akaltara.**
 - **Chhattisgarh** - Notable plants are **Jamul, Durg, and Mandhar.**
 - **Andhra Pradesh** - Cement production is significant in **Vijayawada, Krishna, Guntur, Kurnool, and Vishakhapatnam.**
 - **Telangana** - The plant in **Adilabad** contributes to the state's production.
 - **Rajasthan** - Key locations include **Sawai Madhopur, Udaipur, Lakheri, Churu, and Chittorgarh.**
 - **Gujarat** - Prominent plants are in **Porbandar, Jamnagar, Dwarika, Ahmedabad, Sikka, Vadodara, Bhavnagar, and Veraval.**
 - **Tamil Nadu** - Important sites are **Tulukapatti, Thalaiyuthu, Dalmiapuram, and Tirunelveli.**
 - **Karnataka** - Major plants include **Bhadravati, Shahabad, Kurkunta, Bagalkot, and Gulbarga.**
 - **Maharashtra** - Cement is produced in **Chanda, Ratnagiri, Sevari, and Manikgarh.**
 - **Bihar** - All cement plants are established along the banks of the **Son River**, with major plants in **Banjari, Dalmianagar, and Kalyanpur.**
 - **Jharkhand** - Key plants include **Sindri, Khelari, Jhinkpani** (Chaibasa), and **Japla** (Palamau).
 - **Uttar Pradesh** - Notable locations are **Churk, Dalla, and Chunar.**
 - **Haryana** - Cement production occurs in **Surajpur and Charkhi Dadri.**
 - **Odisha** - Key plants are in **Rajgangpur and Hirakud.**
 - **Himachal Pradesh** - Notable plants include **Rajban and Gagal.**
 - **Jammu & Kashmir** - The plants in **Wuyan and Basoli** are significant for the region.
 - **Assam** - The plant at **Bokajan** contributes to local production.
 - **West Bengal** - Cement is produced in **Durgapur.**
 - **Uttarakhand** - The cement plant in **Dehradun** is also notable.

Chemical Fertilizers Industry:

- India, being predominantly an agricultural country, has witnessed significant growth in the **chemical fertilizers industry** due to the increasing demand for fertilizers. The three main types of chemical fertilizers produced in India are **nitrogenous**, **phosphatic**, and those containing **potash**. Among these, the demand and production of **nitrogenous fertilizers** are particularly high due to the **nitrogen deficiency** in the alluvial soil prevalent in many regions. In contrast, the **plateau regions** have seen a rise in the demand and production of phosphatic and potash fertilizers, corresponding to agricultural growth in those areas.
- The first fertilizer plant in India was established in **Ranipet, Tamil Nadu, in 1906**. This was followed by the establishment of several other plants: the second in **Belegula, Karnataka (1943)**, the third in **Alwaye, Kerala (1947)**, and the fourth in **Sindri, Jharkhand (1951)**, which was notable as the **first public sector fertilizer plant** in the country.
- The localization of the chemical fertilizers industry** is significantly influenced by the availability of **raw materials**. The key raw materials used in this industry include **naphtha**, **coke**, **coke-oven gas**, **electrolytic hydrogen**, **sulphuric acid**, **ammonium sulphate** (derived from gypsum), **natural gas**, and **rock phosphate**. The types of plants based on these materials are as follows:

- Naphtha-Based Plants:**

- **Naphtha**, a **petroleum product** derived from **oil refineries**, is crucial for nitrogenous fertilizer production. The naphtha-based plants are primarily located near oil refineries or seaports:
- **Plants Near Oil Refineries** - Notable examples include **Barauni** (now closed), **Guwahati**, and **Vadodara** (where naphtha is sourced from the Koyali oil refinery).
- **Plants Near Seaports** - Naphtha-based plants are present at almost all major seaports except **Tuticorin**. Approximately **70%** of the country's nitrogenous fertilizer production utilizes naphtha.



- Coal (Coke)-Based Plants:**

- These plants utilize **coal (coke)** as a primary raw material and are located in areas like **Talcher (Odisha)**, **Ramagundam (Telangana)**, **Korba (Chhattisgarh)**, and **Neyveli (Tamil Nadu)**, with the Neyveli plant specifically relying on **lignite coal**.

- **Coke-Oven Gas-Based Plants:**

- Plants that utilize **coke-oven gas** include locations in **Sindri (Jharkhand)**, **Jamshedpur (Jharkhand)**, **Rourkela (Odisha)**, **Bhilai (Chhattisgarh)**, and **Durgapur (West Bengal)**.

- **Electrolytic Hydrogen-Based Plants:**

- **Electrolytic hydrogen** is produced at hydropower plants, and fertilizer facilities at **Nangal (Punjab)** and **Hirakud (Odisha)** are examples of this type.

- **Sulphuric Acid-Based Plants:**

- **Sulphuric acid** is sourced from various origins, including:
- **Oil Refineries** - A plant established in **Kochi**.
- **Zinc Smelting Plants** - A facility located in **Khetri (Rajasthan)**.
- **Pyrite Factories** - A plant in **Alwaye (Kerala)**.

- **Ammonium Sulphate (Gypsum)-Based Plants:**

- Plants utilizing **ammonium sulphate**, derived from gypsum, are located in **Sindri** and **Udaipur**.

- **Natural Gas-Based Plants:**

- Numerous fertilizer plants are based on **natural gas**, transported via the **Hazira-Vijaypur-Jagdishpur (HVJ) pipeline**. Six notable plants include:
- **Vijaypur (Madhya Pradesh)**
- **Sawai Madhopur (Vilopa village, Rajasthan)**
- **Jagdishpur, Shahjahanpur, Aonla, and Babrala (all in Uttar Pradesh)**.

- **Rock Phosphate-Based Plants:**

- These plants are situated in **Khetri (Rajasthan)** and **Udaipur (Rajasthan)**.

Public Sector Undertakings:

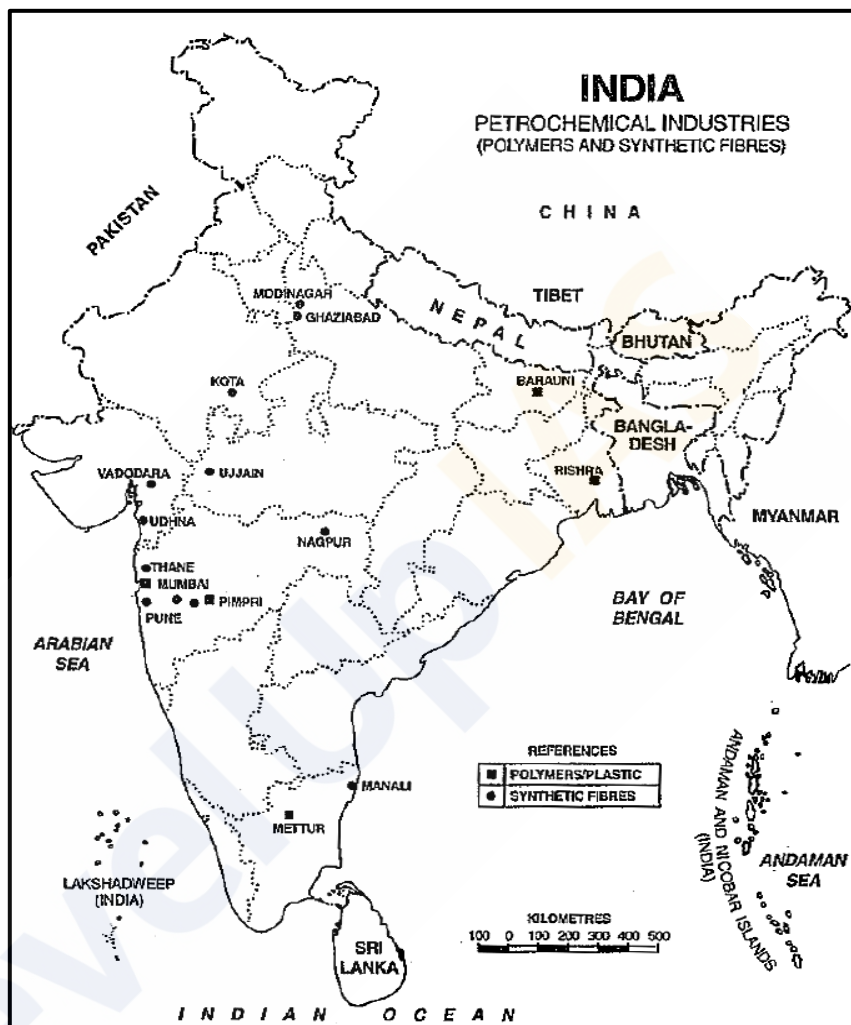
- **Fertilizers Corporation of India Limited (FCI)**, established in **1961**, with plants in **Sindri (Jharkhand)**, **Gorakhpur (Uttar Pradesh)**, **Talcher (Odisha)**, and **Ramagundam (Telangana)**.
- **National Fertilizers Limited (NFL)**, founded in **1974**, operating in **Nangal (Punjab)**, **Bhatinda (Punjab)**, **Panipat (Haryana)**, and **Vijaypur (Madhya Pradesh)**.
- **Hindustan Fertilizers Corporation Limited (HFC)**, initiated in **1978**, with plants in **Namrup (Assam)**, **Durgapur (West Bengal)**, and **Barauni (Bihar)**.
- **Rashtriya Chemicals and Fertilizers Limited (RCF)**, also established in **1978**, with gas-based plants in **Trombay** and **Thal (Maharashtra)**.
- **Fertilizers and Chemicals Travancore Limited (FACT)**, operating in **Udyogamandal (Kerala)** and **Kochi (Kerala)**.
- **Pyrites, Phosphates and Chemicals Limited (PPCL)**, with facilities in **Amjhor (Bihar)** and **Saladipura (Rajasthan)**.
- **Paradeep Phosphates Limited (PPL)**, located in **Paradeep (Odisha)**.
- **Indian Farmers Fertilizer Co-operative Limited (IFFCO)**, with plants in **Kalol (Gujarat)**, **Kandla (Gujarat)**, **Phoolpur (Uttar Pradesh)**, and **Aonla (Uttar Pradesh)**.
- **Krishak Bharti Co-operative Limited (KRIBHCO)**, located in **Hazira**.

Petrochemical Industry:

- The **petrochemical industry** plays a crucial role in transforming crude petroleum into various products that serve as raw materials for numerous other industries. The products derived from crude oil are classified into four primary categories: **polymers**, **artificial fibers**, **illustromers**, and **surfactant intermediates**.

- **Polymers:**

- Polymers are synthesized from **ethylene** and **propylene**, which are byproducts obtained during the refining of crude oil. These materials are fundamental to the **plastic industry**, as they serve as the building blocks for a wide array of plastic products. A notable pioneer in this sector is the **National Organic Chemical Industry Limited (NOCIL)**, recognized as the country's first naphtha-based chemical industry. Established in **1961** in **Mumbai** by the **Mafatlal Group**, NOCIL has been instrumental in the growth of polymer production in India. Other major centers for plastic production include **Barauni (Bihar)**, **Mettur (Tamil Nadu)**, **Pimpri (Pune)**, and **Rishra (West Bengal)**.



- **Artificial Fibers:**

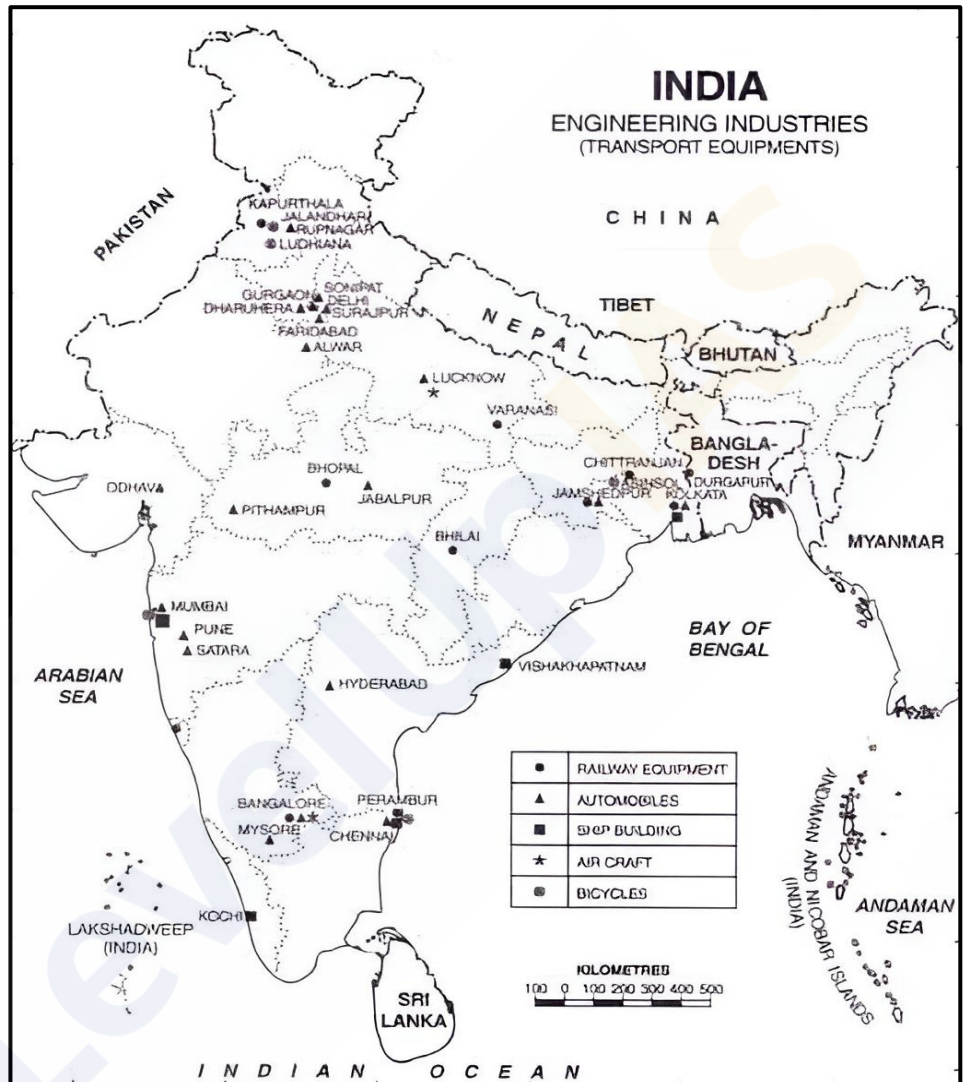
- Artificial fibers are extensively utilized in the **textile industry**, with **nylon** and **polyester** being the predominant types. The surge in this industry can be traced back to the **1960s**, when raw materials derived from petroleum refineries significantly boosted production capabilities. Manufacturing plants for nylon and polyester fibers have been established in various locations, including **Kota**, **Pimpri**, **Mumbai**, **Modinagar**, **Pune**, **Ujjain**, **Nagpur**, and **Udhana**. Additionally, **acrylic staple fibers** are produced in **Kota** and **Vadodara**, while **polyester staple fibers** can be found in **Thane**, **Ghaziabad**, **Manali**, **Kota**, and **Vadodara**.
- In addition to polymers and artificial fibers, the production of **illustromers** and **artificial detergents** has also become prominent, particularly at the **Haldia Petrochemical Industry**. This facility exemplifies the diverse applications of petrochemicals beyond traditional uses.
- The manufacture of explosives is concentrated in several key regions, including **Auraiya (Uttar Pradesh)**, **Jamnagar**, **Gandhinagar**, **Hazira (Gujarat)**, **Nagothane**, **Ratnagiri (Maharashtra)**, **Sivakasi (Tamil Nadu)**, and **Vishakhapatnam (Andhra Pradesh)**. The **Haldia** region in **West Bengal** is also notable for its involvement in explosives production.

Engineering Industry:

- The engineering industry in India has witnessed significant development since its inception, particularly in the **heavy engineering** sector, **machine tools**, **heavy electric machinery**, **electronic industry**, **information technology**, and **rail equipment**. Below is a structured overview of these sectors, highlighting key developments and notable companies.

Heavy Engineering Industry:

- The **heavy engineering industry** in India was established in **1958** with the formation of the **Heavy Engineering Corporation Limited (HEC)** in Ranchi, Jharkhand, with assistance from **Russia** and **Czechoslovakia**. Key milestones in this sector include:
 - Triveni Structurals Limited**, established in **1965** in Naini (Allahabad) with Austrian collaboration, focuses on creating specialized steel frames.
 - Tungbhadra Steel Products Limited** in Tungbhadra, Karnataka, stands out as a significant unit for heavy engineering production.
 - Bharat Heavy Plate and Vessels Limited**, initiated in **1966** in Vishakhapatnam with Czechoslovak assistance, manufactures machinery for the **fertilizers**, **petrochemicals**, and other related industries.
 - Bharat Pumps and Compressors Limited** was also established in Naini (Allahabad), contributing to pump and compressor manufacturing.
 - Mining and Allied Machinery Corporation Limited (MAMC)** was set up in Durgapur, specializing in mining machinery.
 - TEXMACO**, a private sector enterprise founded in **1939** in Mumbai, manufactures textile machinery.
 - Additionally, the **Heavy Machine Tools Plant** in Ranchi produces axles and wheels for heavy machinery.



Machine Tools:

- In the realm of **machine tools**, several plants have been established across the country. **Hindustan Machine Tools (HMT)**, founded in **1953** in Bengaluru with Swiss assistance, is one of the leading public sector undertakings. HMT operates additional units in **Pinjore** (Haryana), **Kalamassery** (Kerala), **Hyderabad** (Telangana), **Ajmer** (Rajasthan) and **Srinagar** (Jammu and Kashmir).

- HMT specializes in the manufacturing of **tractors, watches, and industrial machines**. However, in **September 2014**, the Government of India announced the phased shutdown of HMT's watch plants.

Heavy Electric Machinery:

- The **Bharat Heavy Electricals Limited (BHEL)** plays a pivotal role in the **heavy electric machinery** sector. Established as a public sector entity, BHEL's plants are strategically located in **Bhopal, Tiruchirapalli, Hyderabad, Haridwar, Ranipet, Bengaluru** and **Jamshedpur**.
- These facilities focus on manufacturing electric machinery to support various industrial needs.

Electronic Industry:

- The **electronic industry** in India has its roots in the establishment of the **Indian Telephone Industries** in **1950** in Bengaluru, with additional units in **Srinagar, Raebareli, and Naini** (Allahabad). Another notable player is **Bharat Electronics Limited (BEL)**, founded in the 1950s in Bengaluru, which produces electronic equipment for **military, radio, and meteorological** applications. **Hindustan Cables** is also significant in this sector, manufacturing telephone wires with plants in **Roopnarayanpur** (Burdwan district, West Bengal) and **Hyderabad**.

Information Technology Industry:

- The **information technology industry** encompasses a wide range of fields, including **electronic communication, telecommunication, radio, television, and computing**. **Bengaluru** is recognized as the major hub for this sector, often referred to as the **Silicon Valley of India** due to its concentration of tech companies and talent.

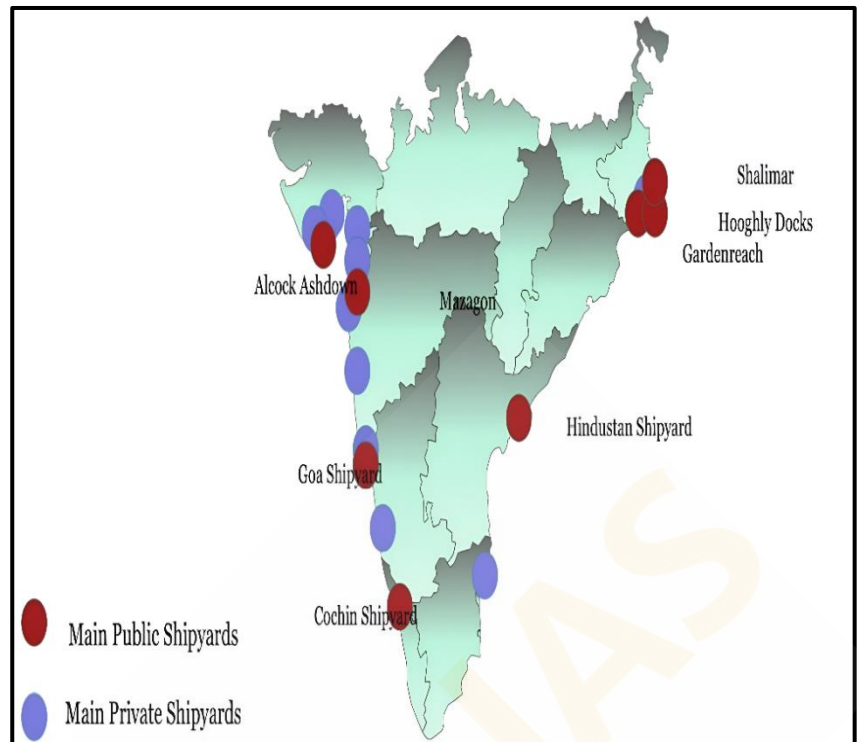
Rail Equipment:

- The history of rail equipment manufacturing in India dates back to the **late 19th century**, with locomotive factories established in **Jamalpur** (Bihar) and **Ajmer** (Rajasthan) that were eventually closed due to lack of support from the British government. Currently, India has two major locomotive factories:
 - **Chitranjan Locomotive Works**, established in **1950** in Chitranjan (West Bengal), specializes in manufacturing electric locomotives.
 - **Diesel Locomotive Works**, located in Maduadih (Varanasi), was set up in **1961** with assistance from the American company **Alco**, focusing on diesel locomotives.
 - **Tata Engineering and Locomotive Works (TELCO)** in Jamshedpur, which produced steam locomotives until 1970.
 - **Diesel Components Works** in Patiala (Punjab), specializing in components for diesel locomotives.
 - The **Integral Coach Factory** in Perambur (Tamil Nadu), established in **1955** with Swiss assistance, produces general and air-conditioned coaches.
 - **Rail Coach Factory** in Kapurthala (Punjab) manufactures rail coaches, while **Messers Jessops** in Kolkata and **Bharat Earth Movers Limited** in Bengaluru also produce passenger coaches.
- Plans are underway for new facilities, including a coach wheel manufacturing plant in **Chapra, Bihar**, and a rail coach factory in **Raebareli, UP**, although land acquisition disputes have stalled progress.

Ship Building Industry:

- The **shipbuilding industry** in India has a rich history, with the establishment of its first technologically advanced shipbuilding plant in **1941**. This plant was set up in **Vishakhapatnam** by the **Scindia Steam Navigation Company**. The first vessel constructed at this facility was named '**Jal-Usha**.' In **1952**, the shipyard was nationalized and renamed **Hindustan Shipyard Limited**, becoming a significant player in the Indian maritime sector.

Major Shipyards in India:



- Hindustan Shipyard Limited, Vishakhapatnam:**
 - Hindustan Shipyard Limited is notable for its capability to construct **high-capacity cargo ships**. Located in Vishakhapatnam, this shipyard plays a crucial role in meeting the demand for commercial maritime transportation.
- Garden Reach Shipbuilders and Engineers Limited, Kolkata:**
 - Situated on the banks of the **Hugli River**, the **Garden Reach Shipbuilders and Engineers Limited** specializes in building **passenger ships, dredgers, and coasters**. This shipyard is known for its contribution to the coastal and inland waterway transportation sector.
- Mazagaon Dock, Mumbai:**
 - The **Mazagaon Dock** in Mumbai is primarily focused on the construction of **naval ships**, including **frigates**. It plays a vital role in supporting India's defense capabilities by building advanced naval vessels.
- Cochin Shipyard Limited, Kochi (Kerala):**
 - The **Cochin Shipyard Limited** is the largest shipyard in India and was developed with the assistance of **Japan**. This facility not only focuses on shipbuilding but also engages in **repairing ships** and providing **marine engineering training**. It is a comprehensive hub for maritime activities.
- Goa Shipyard Limited, Vasco da Gama, Goa:**
 - Located in Vasco da Gama, Goa, the **Goa Shipyard Limited** specializes in building and repairing ships for the **navy, coast guard, and civil purposes**. This shipyard contributes to the nation's maritime security and civil maritime operations.

Aircraft Manufacturing Industry:

- India's first major step towards developing its own **aircraft manufacturing industry** took place in 1940 with the establishment of **Hindustan Aircraft Limited (HAL)** in **Bengaluru**. During the same period, aircraft manufacturing activities were also carried out at the **Aircraft Manufacturing Depot** in **Kanpur**. In **1964**, these two entities were merged to form a unified company, which was renamed **Hindustan Aeronautics Limited (HAL)**, with its headquarters in **Bengaluru**.

HAL's Manufacturing Units and Operations:

- HAL has expanded its operations and now operates **6 major units** across various locations in India. Each unit specializes in specific areas of aircraft manufacturing and defence technology:
 - Bengaluru** - This unit focuses on **aircraft assembly**. It assembles the **Jaguar aircraft** and manufactures **engines** for **Chetak** and **Cheetah helicopters**.
 - Kanpur** - The unit in Kanpur specializes in the production of **aircraft chassis**. Notably, it manufactures the chassis for the **Dornier aircraft**, primarily for the **Indian Navy**, with technical assistance from **Germany**.
 - Nasik** - Similar to Kanpur, the **Nasik** unit is involved in **aircraft chassis** manufacturing, contributing significantly to India's defence capabilities.
 - Koraput, Odisha** - This unit is dedicated to the production of **aircraft engines**. A notable development at this facility is a specialized **RADAR system** designed for the **Light Combat Aircraft (LCA)**.
 - Hyderabad** - This unit focuses on manufacturing **spare parts** and **electronic equipment** related to **aircraft communication systems**, ensuring smooth and reliable operations of military aircraft.
 - Korwa, Lucknow** - The Korwa unit is responsible for producing **specialized electric and communication equipment** for both the **Indian Air Force** and the **Navy**. It is also known for developing a specific type of integrated equipment called **DARIN** (Display Attack Ranging Inertial Navigation), which is utilized in the **Light Combat Aircraft (LCA)**.

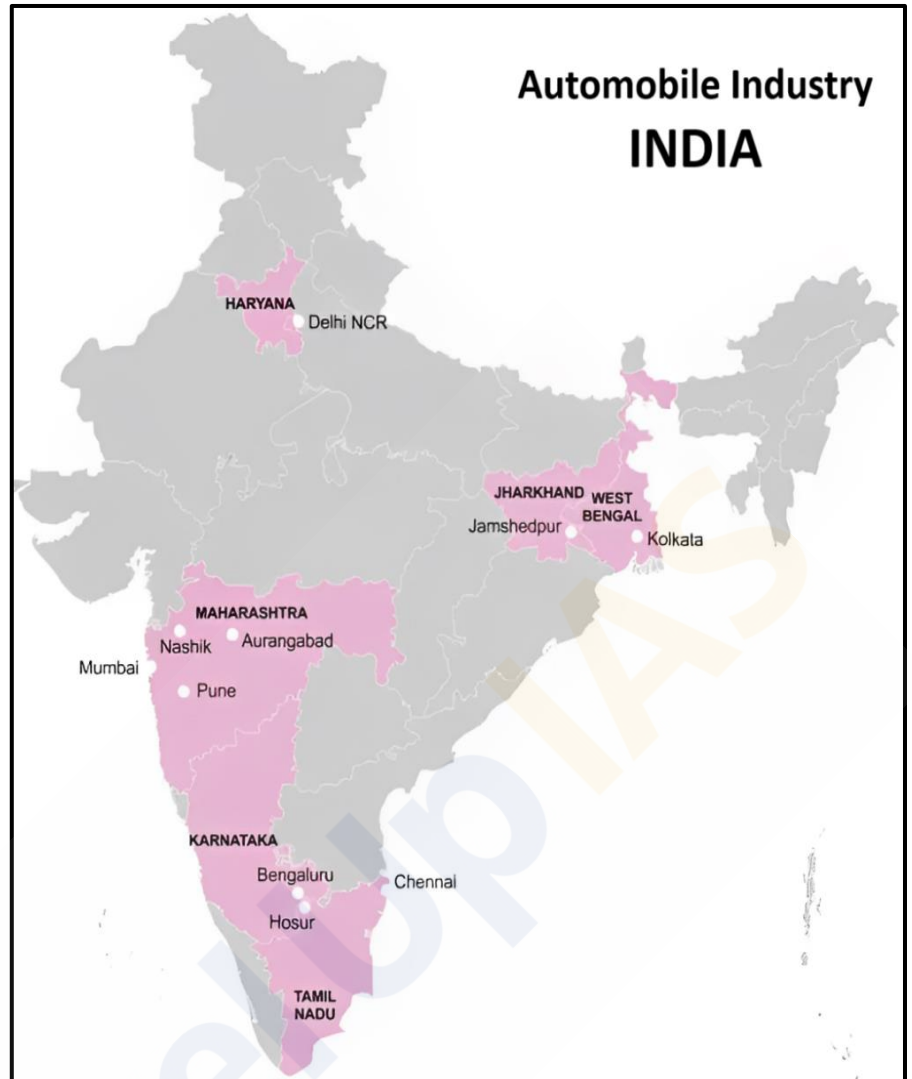
Automobile Industry:

- The **automobile industry in India** began in **1928** when the **General Motors Company** was set up in **Mumbai**. Shortly after, in **1930**, the **Ford Company** established a plant in **Chennai**, marking the early stages of the Indian automobile sector. Later, during **1944**, significant developments took place with the establishment of **Hindustan Motors Limited** at **Uttarpara, Kolkata**, and **Premier Automobiles Limited** in **Kurla, Mumbai**. Initially, vehicle production in India relied heavily on assembling **imported components**. However, following **Independence**, the government imposed **high taxes on imported spare parts** to promote indigenous production, a move that played a crucial role in boosting the domestic manufacturing industry.

Major Automobile Manufacturing Centers in India:

- Hindustan Motors**, based in **Uttarpara, Kolkata**, was once famous for producing the iconic **Ambassador cars**, although the company is now **defunct**.
- Premier Automobiles Ltd** in **Mumbai** became well-known for manufacturing the **Padmini (Fiat)**.

- **Standard Motor Production India Ltd** established its operations in **Chennai**.
- **Ashok Leyland Ltd**, also located in **Chennai**, has become a key player in manufacturing commercial vehicles.
- **Tata Engineering and Locomotives Company (TELCO)** is based in **Jamshedpur**, known for its wide range of vehicles.
- **Mahindra and Mahindra Ltd** operates out of **Pune**, specializing in utility vehicles and tractors.
- **Maruti Udyog Ltd**, headquartered in **Gurugram, Haryana**, is a leading car manufacturer. The company also has major manufacturing units in **Manesar, Haryana (established in 2007)** and **Hansalpur, Gujarat (established in 2017)**.



- The **Tata Nano Car** is manufactured at the **Sanand plant in Gujarat**, a project that drew significant attention due to its focus on low-cost cars.
- The **Volkswagen Group**, a leading German car manufacturer, set up its production facility in **Chakan, Pune**.
- **Skoda**, a Czech automobile manufacturer, has established units in **Chakan (Pune)** and **Aurangabad, Maharashtra**.

Two-Wheeler and Tractor Manufacturing:

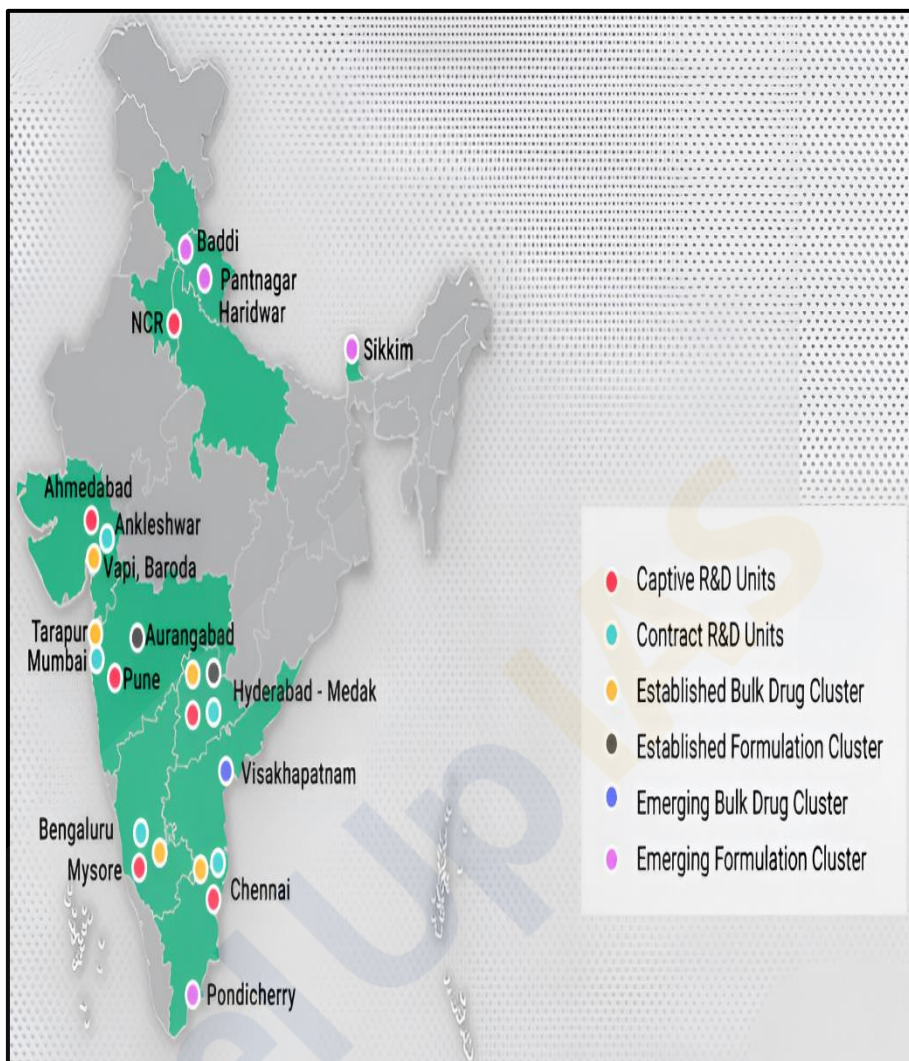
- India also has a robust manufacturing sector for **motorcycles and two-wheelers**. Major production hubs include **Pune, Mumbai, Faridabad, Chennai, Kanpur, Mysore, and Ludhiana**.
- Additionally, **Faridabad (Haryana), Pinjore (Haryana), Delhi, Mumbai, and Chennai** are prominent centers for the **production of tractors**, which play an essential role in India's agricultural sector.

Bicycle Manufacturing Hubs:

- The production of bicycles is a well-established industry in India. Key centers include **Sonipat (Haryana), Jalandhar (Punjab), Ludhiana (Punjab), Mumbai, and Asansol (West Bengal)**.
- These hubs contribute significantly to domestic transportation and provide an affordable mode of travel for millions.

Pharmaceutical Industry:

- The **Indian Drugs and Pharmaceuticals Ltd. (IDPL)** is one of the largest and most prominent pharmaceutical companies in India. It plays a significant role in the production of essential drugs and pharmaceuticals for the country. IDPL has established several manufacturing plants located across different regions of the country such as **Rishikesh** (Uttarakhand), **Hyderabad** (Telangana), **Gurgaon** (Haryana), **Chennai** (Tamil Nadu) and **Muzaffarpur** (Bihar).
- Another key player in India's pharmaceutical sector is **Hindustan Antibiotics Ltd. (HAL)**, which was set up in **Pimpri (Pune)**. HAL specializes in the production of antibiotics such as **penicillin**, **streptomycin**, and other important pharmaceutical products used in healthcare.
- The **Bengal Chemicals and Pharmaceuticals Ltd. (BCPL)**, founded in 1981, is a significant contributor to India's pharmaceutical production. BCPL operates several units in various cities, including **Maniktala** (West Bengal), **Panihati** (West Bengal), **Mumbai** (Maharashtra), and **Kanpur** (Uttar Pradesh)
- **Hindustan Organic Chemicals Ltd. (HOCL)**, located in **Rasayani (Maharashtra)**, is another key organization involved in the pharmaceutical industry. Besides the production of medicines, HOCL manufactures **biochemicals** used in industries such as **dye stuffs** and **plastics**. A separate plant of HOCL in **Kochi (Kerala)** produces **phenol**, a crucial chemical in many industrial processes.



Insecticides and Pesticides Industry:

- In the realm of insecticides and pesticides, **Hindustan Insecticides Ltd.** is a leading entity. The company has key units located in **New Delhi** (where **DDT** is produced) and **Alwaye, Kerala** (which focuses on the production of **BHC** (Benzene Hexachloride), another vital pesticide used in agriculture).
- In addition to its pharmaceutical contributions, **Hindustan Organic Chemicals Ltd. (HOCL)**, which operates out of Rasayani, also has an important presence in the production of chemicals for the insecticides and pesticides industry. Its **Kochi** plant is responsible for manufacturing **phenol**, a crucial input for several chemical processes.
- **Hindustan Fluorocarbons Ltd.**, located in **Medak (Telangana)**, is an associate company of HOCL and is involved in the production of various chemical products, further enhancing the chemical industry's capabilities.

Research and Education in Pharmaceuticals:

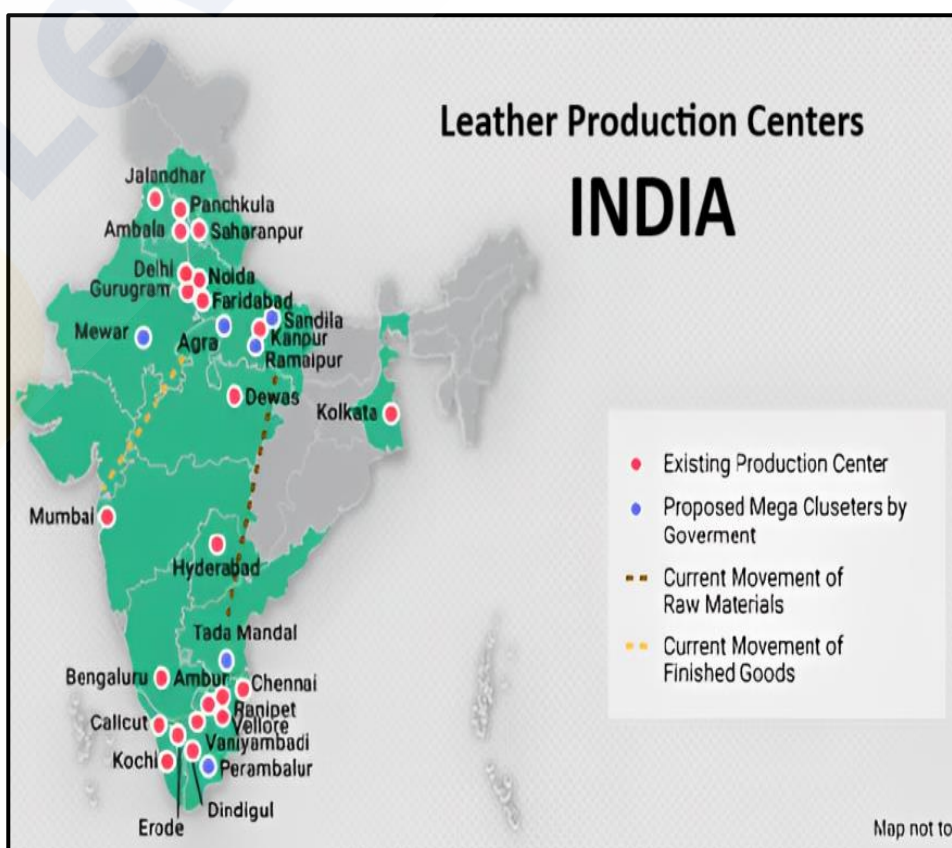
- To further advance research and training in the pharmaceutical industry, the **National Institute of Pharmaceuticals Education and Research (NIPER)** was established in **Mohali (Chandigarh)**. This institute is a premier center for fostering innovation and excellence in pharmaceutical education and research in India.

Glass Industry:

- The **glass industry in India** is characterized by both **cottage-scale** and **large-scale factory production**. While traditional craftsmanship continues to thrive in smaller cottage industries, modern glass manufacturing is also carried out on a large scale across various regions of the country.
- One of the most significant hubs of the cottage industry for glass production is **Firozabad**, located in Uttar Pradesh. Firozabad is particularly renowned for the **manufacturing of glass bangles**, making it a pivotal center in the glass industry. Other major centers for glass production in Uttar Pradesh include **Shikohabad**, **Bahzoi (Moradabad)**, **Naini (Allahabad)**, and **Hathras**. **Glass sheets** are primarily produced in **Bahzoi and Naini**.
- In **West Bengal**, glass manufacturing is concentrated in areas like **Kolkata**, **Belur**, **Rishra**, **Raniganj**, **Asansol**, **Dum Dum**, **Belgachhia**, and **Belgharia**. These areas contribute significantly to the region's output of glass products.
- Maharashtra** is another major center, with production facilities in **Mumbai**, **Talegaon (Pune)**, **Orgalewadi (Satara)**, **Kolhapur**, and **Nagpur**. This state is known for the manufacturing of **bottles**, **flasks**, **beakers**, **lamps**, and **glass sheets**.
- Gujarat** has its centers of glass production in **Bharuch**, **Vadodara**, and **Panch Mahal**, while **Tamil Nadu** hosts glass industries in **Salem**, **Chennai**, and **Coimbatore**. In **Rajasthan**, glass is produced in **Dhaulpur** and **Jaipur**, while **Kerala** has a notable glass manufacturing center in **Alwaye**.
- Other important centers across India include **Shahdara (Delhi)**, **Ambala and Faridabad (Haryana)**, and **Jabalpur (Madhya Pradesh)**.

Leather Industry:

- India, with its vast livestock resources, has a well-developed **leather industry**, making it one of the leading producers globally. The industry has flourished due to the abundant availability of raw hides, particularly in the cities of **Kolkata**, **Kanpur**, **Chennai**, and **Coimbatore**, which serve as major markets for hides.
- India is home to numerous **tanneries**, with key centers in **Kolkata**, **Chennai**, **Mumbai**, **Agra**, **Tonk**, **Bengaluru**, **Fulbani**, **Belgaum**, **Perambur**,



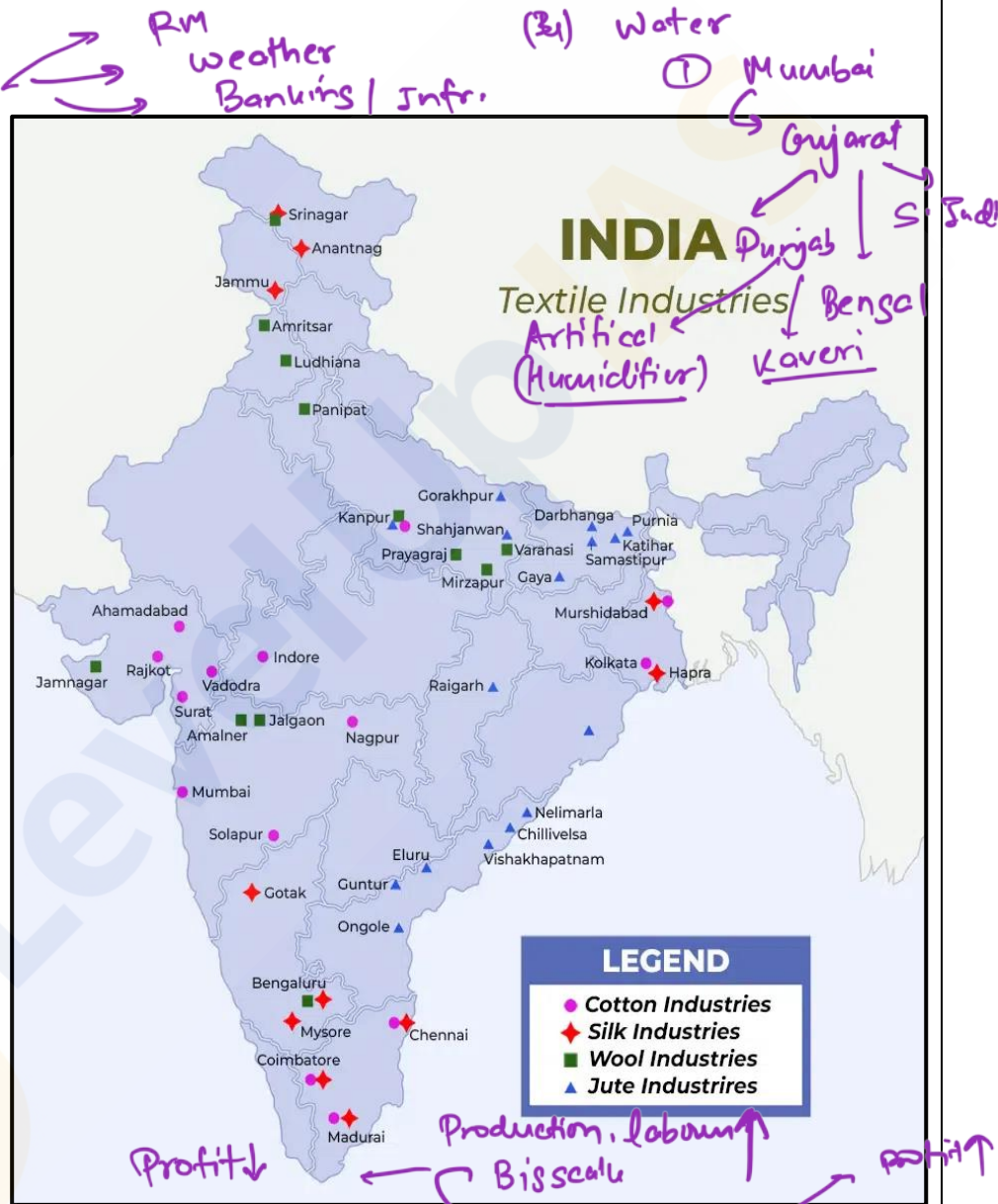
Tiruchirapalli, and Sherbag (Gujarat). These tanneries play a crucial role in processing hides into leather, which is then used for manufacturing various products.

- When it comes to **Shoe production**, several cities in India stand out as major hubs. The prominent centers include **Batanagar (Kolkata), Chennai, Ambur, Ranipet, Dindigul (Tamil Nadu), Agra, and Kanpur (Uttar Pradesh), as well as Dewas (Madhya Pradesh).**
- India's **leather products** are primarily exported from **Chennai harbour**, with other significant export points being **Kolkata and Mumbai**. The leather industry not only caters to the domestic market but also plays a vital role in India's export economy, contributing to the global supply of leather goods.

Agro-Based Industries:

Cotton Textile Industry:

- The **Cotton Textile Industry** has played a significant role in India's industrial development. The first modern cotton textile mill was established in **1818 at Fort Gloster, Kolkata**, but it was unsuccessful. The breakthrough for the Indian cotton textile industry came in **1854**, when **Kavasji Davar** set up the first successful mill using Indian capital in **Mumbai**. This industry relies heavily on the availability of raw materials and can thus be established either near the source of raw materials or closer to markets. Initially, the industry developed in cotton-producing areas such as **Mumbai, Nagpur, and Ahmedabad**. However, over time, it expanded and **decentralized** into regions with better market access, such as **Kanpur**.



- The cotton textile industry in India operates across three main sectors: the **mill sector**, the **powerloom sector**, and the **handloom sector**. Over the years, the contribution of these sectors to the overall production of cotton textiles has shifted significantly. In the **1950s**, the **mill sector** contributed a dominant **81%** of total production, but this share declined sharply to **about 4% by 2019-20**. In contrast, the **powerloom sector** saw substantial growth during the same period, producing around **60% of India's cotton textiles by 2019-20**. The **handloom sector** also maintained a notable presence, contributing approximately **15%** to the total production of cotton textiles in **2019-20**.
- The cotton textile industry has grown significantly over the years. Between **1950-51 and 2014-15**, there was an approximately **8.77 times increase in production**, highlighting the industry's vast expansion and importance in India's economy.

Distribution of Cotton Textile Industry in India:

- The cotton textile industry holds a significant place in India's industrial sector, with several states serving as major hubs due to favourable conditions. The industry's growth is driven by various geographical, climatic, and infrastructural advantages in different regions of the country.

Maharashtra:

- Maharashtra holds a **foremost position in India's cotton textile industry**, housing **190 cotton textile mills**. Mumbai, the state's capital, is the major center of this industry and is often referred to as the '**Cottonopolis of India**', with **57 mills** operating in the city. Apart from Mumbai, Pune, Solapur, Kolhapur, Nagpur, Wardha, Sangli, and other cities are notable centers of cotton textile production in the state.
- **Major Reasons for the Development of the Cotton Textile Industry in Maharashtra:**
 - Marine (humid) climate, which is favourable for cotton production.
 - Availability of local cotton grown in the black cotton soil.
 - Hydroelectric power supply, essential for the operation of mills.
 - Access to Mumbai harbour, facilitating easy import of machinery and raw materials and export of finished goods.
 - Abundant availability of cheap labor.

Gujarat:

- Gujarat ranks **second in India's cotton textile industry**, with **89 mills**. The city of **Ahmedabad** is the principal center, often referred to as the '**Manchester of India**' or the '**Boston of the East**', with **47 mills**. This nickname reflects the city's prominence, similar to Manchester (Great Britain) and Boston (USA), both recognized globally for their textile industries. Apart from Ahmedabad, other important centers include **Surat, Bharuch, Rajkot, Vadodara, Bhavnagar, and Porbandar**. Surat is particularly famous for its **zari (gold and silver embroidery) work**.
- **Major Reasons for the Development of the Cotton Textile Industry in Gujarat:**
 - Marine (humid) climate, favourable for cotton processing.
 - Local supply of cotton grown in black cotton soil.
 - Hydroelectric power availability.
 - Access to Kandla harbour for imports and exports.
 - Availability of cheap labor.

Tamil Nadu:

- Tamil Nadu **leads in terms of the number of cotton textile mills** in India, with **959 mills**. The state accounts for **38% of the country's mill-spun cotton** production. **Coimbatore**, also known as the '**Manchester of South India**', is the major center, with **81 mills**—the highest number of mills in any single Indian city. Other prominent centers include **Chennai, Madurai, Tirunelveli, Tuticorin, Salem, Tanjavur, and Ramanathapuram**.
- **Major Reasons for the Development of the Cotton Textile Industry in Tamil Nadu:**
 - Cheap hydroelectric power, particularly from the Paikara project supplying electricity to Coimbatore and other cities.
 - Local cotton production.
 - Availability of cheap labor.
 - A strong internal market for cotton textiles.

West Bengal:

- In West Bengal, the cotton textile industry is centered around **Kolkata**, with **30 mills** operating in the state.
- **Major Reasons for the Development of the Cotton Textile Industry in West Bengal:**
 - **Easy import-export access via Kolkata harbour.**
 - **Supply of cheap electricity** from the Damodar Valley Corporation (DVC).
 - **Cheap labor**, especially from neighboring states like Bihar, Jharkhand, and Assam.
 - **Large local market.**
 - **Humid climate**, beneficial for cotton processing.

Uttar Pradesh:

- Uttar Pradesh houses **66 cotton textile mills**, with **Kanpur** being the most prominent center. It is often called the '**Manchester of North India**' and has **14 mills**. Other major centers in the state include **Modinagar, Hathras, Saharanpur, Agra, and Lucknow**.
- **Major Reasons for the Development of the Cotton Textile Industry in Uttar Pradesh:**
 - **Large market** for cotton textiles.
 - **Availability of cheap labor.**

Karnataka:

- In Karnataka, the cotton textile industry is concentrated in the **eastern parts**, where cotton is widely produced. The major centers include **Devnagari, Hubli, Bellary, Mysore, and Bengaluru**.

Telangana:

- In Telangana, the cotton textile industry has developed in regions where cotton is grown. Most mills in the state are **spinning mills**, producing yarn. Major centers include **Hyderabad, Secunderabad, and Warangal**.

Punjab:

- Punjab has witnessed a rapid increase in cotton textile production, with **90 mills** operating in the state due to rising cotton production. Major centers include **Amritsar, Ludhiana, and Phagwara**.

Problems Faced by Cotton Textile Industry in India:

- **Lack of High-Quality Raw Material** - One of the primary issues is the **limited availability of high-quality raw materials**, particularly **long staple cotton**. This type of cotton is crucial for producing finer yarns and higher-grade fabrics, but India has a shortage of it. As a result, the industry often relies on imports, increasing production costs and reducing its competitiveness.
- **Inadequate Infrastructural Facilities** - The industry suffers from **poor infrastructure**, which hampers efficient production and distribution. A major problem is the **insufficient power supply**, leading to frequent power cuts and unreliable electricity, especially in smaller, decentralized units. This not only affects production timelines but also increases operational costs due to the need for alternative power sources like generators.
- **Obsolete Machinery and Technology** - Another significant challenge is the **use of outdated machinery and technology**. Many textile mills in India continue to operate with **obsolete machines**, which reduces production efficiency and output quality. The inability to upgrade to modern technology limits the industry's ability to innovate and compete globally.

- **Stiff Competition from Synthetic Textiles** - The rise of synthetic textiles, particularly those made from **mixed and artificial fibers**, poses a serious threat to the cotton textile industry. **Synthetic fabrics are often cheaper and more durable**, making them more appealing to consumers. This shift in consumer preference has reduced the demand for cotton textiles, impacting sales and profitability.
- **Global Competition** - India faces intense competition from other major cotton textile producers, particularly **China**, which dominates the global market. China's **superior technology, economies of scale, and export-friendly policies** give it a significant advantage over India. Despite being the **second-largest producer and exporter** of cotton textiles in the world (as of 2018-2019), India struggles to match China's production capacity and price competitiveness.

Sugar Industry: *Weight lossing*
Assured supply
co-operative sector

Punjab
(Post Green Revolution)
Kovani

- In India, **sugar is primarily produced from sugarcane**, unlike in Europe, where sugar is largely derived from **sugar beets**. The first attempt to produce sugar in India using modern techniques was made by Dutch merchants in **1840 in north-western Bihar**, but it was **unsuccessful**. The first successful sugar mill was established by the British in 1903 at **Marhowrah in the Saran district of Bihar**. Following this, several sugar mills were set up in Bihar and adjacent areas of Uttar Pradesh (UP). By 1931, there were 31 sugar mills in northern India. In 1932, the industry gained **government support**, with a tax imposed on sugar imports, leading to the establishment of 137 sugar mills by 1937.
- The sugar industry is a **weight-losing industry**, meaning the raw material, sugarcane, loses a significant portion of its weight during processing. Typically, **9-12 kg of sugar** is produced from every **100 kg of sugarcane**. Given the bulkiness and perishability of sugarcane, transporting it is difficult compared to sugar, necessitating the **location of sugar mills near sugarcane-producing areas**. Additionally, sugarcane must be **crushed within 24 hours** of harvesting to avoid a reduction in **sucrose content**.

Growth region

Tropical
Sub-Tropical

Variety gives more juice

90% sugar content reduced within 24 hrs.

Major Sugar-Producing States in India:

- The leading sugar-producing states in India are **Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu, Gujarat, Bihar, and Andhra Pradesh**. Below is a breakdown of sugar production in key states:

Uttar Pradesh:

- **Ganga-Yamuna Doab Region** - Major sugar-producing areas include **Saharanpur, Muzaffarnagar, Meerut, and Ghaziabad**.
- **North-Eastern Terai Region** - Important districts are **Gorakhpur, Basti, Deoria, Gonda, Sitapur, and Faizabad**.

Maharashtra:

- Maharashtra is a significant producer of sugarcane and sugar in South India. The state's fertile **black soil** irrigated by the **Godavari, Krishna, Pravara, and Neera rivers** contributes to **high sucrose content** in the sugarcane, leading to higher sugar yields. The **sugar mills are concentrated in a narrow belt** from **Manmad in the north to Kolhapur in the south**. Key sugar production centers include **Ahmednagar, Kolhapur, Solapur, Pune, Satara, Sangli, Nasik, Malinagar, and Sakkarwadi**. Most of these mills operate under the **cooperative sector**, promoting better engagement with farmers.

Tamil Nadu:

- Tamil Nadu has a high **sugarcane yield per hectare** with a longer crushing season (around **160 days**), which enhances sugar production. The use of modern machines further boosts efficiency. **Coimbatore** is the leading district, followed by **Tiruchirapalli, Vellore, and North and South Arcot**.

Karnataka

- Sugar mills in Karnataka are spread across **Belgaum, Bellary, Mandya, Shimoga, Bijapur, and Chitradurga** districts.

Andhra Pradesh:

- In Andhra Pradesh, sugar mills are primarily located in the **East and West Godavari** districts of the coastal region and **Chittur** in the interior **Royalaseema** region.

Telangana

- The main sugar-producing districts in Telangana are **Nizamabad** and **Medak**.

Gujarat

- In Gujarat, sugar production is concentrated in **Surat** and other areas like **Junagarh, Rajkot, Amreli, Valsad, and Bhavnagar**.

Bihar

- Bihar's sugar mills are concentrated in the **north-western districts** like **West and East Champaran, Saran, Muzaffarpur, Gopalganj, Siwan, and Darbhanga**. Other important sugar-producing districts include **Gaya, Bhagalpur, Patna, and Bhojpur**.

Punjab and Haryana:

- Punjab's sugar mills are mainly located in **Gurdaspur, Jalandhar, Sangrur, Patiala, and Amritsar**, while Haryana's key sugar-producing districts include **Yamunanagar, Rohtak, Hissar, and Faridabad**.

Rajasthan:

- In Rajasthan, **Ganganagar** and **Bhupal Sagar** are significant centers for sugar production.



Decentralization of the Sugar Industry:

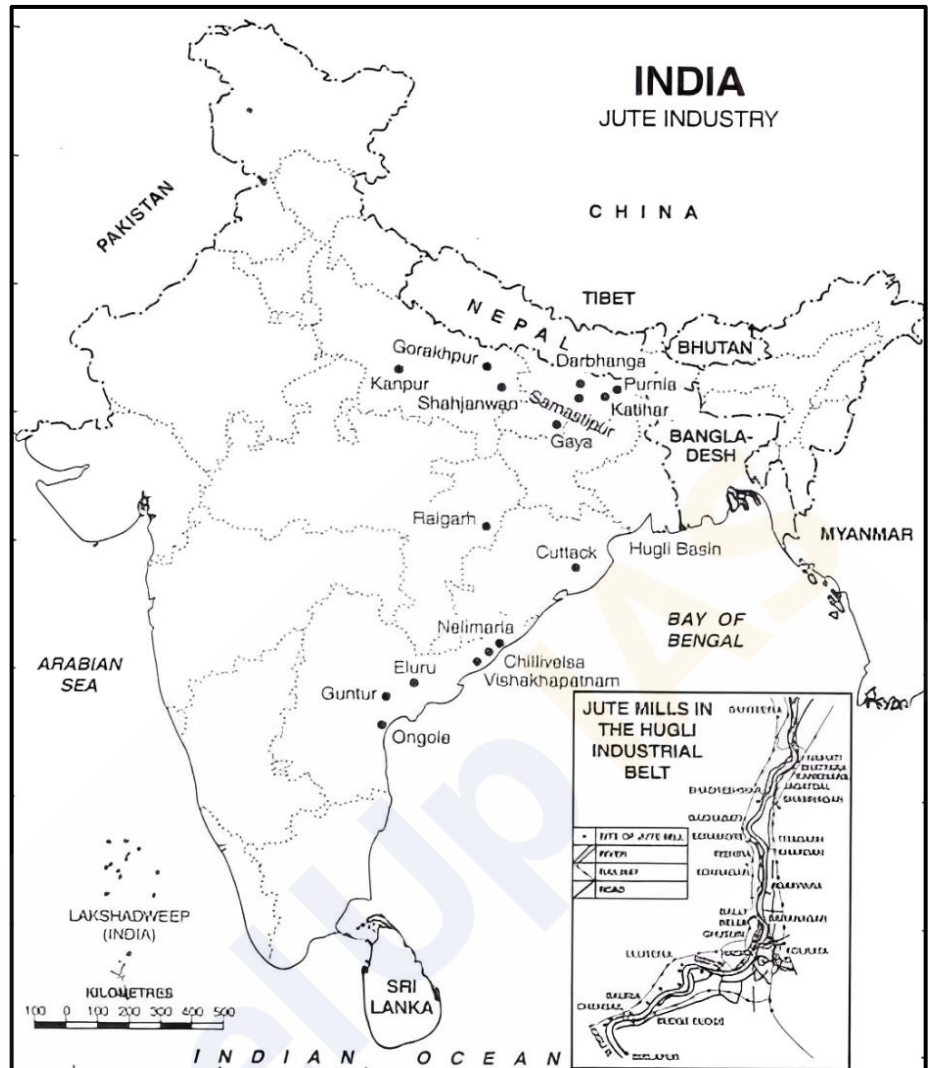
- While the sugar industry initially flourished in **Bihar and Uttar Pradesh**, the **1970s saw decentralization** toward the southern states. **Several factors contributed to this shift such as:**
 - The **black soil of the Deccan plateau** in South India, which has a high moisture-retention capacity and **limestone content**, is ideal for sugarcane cultivation.
 - The **longer crushing period** in southern India due to the **marine humid climate**.
 - **Cooperative ownership** of sugar mills in South India, which has fostered stronger relationships between farmers and the industry.
 - Sugarcane in southern India has **higher juice content**, with a **recovery rate of around 11.6%**.
 - The development of **hydro-electric power** has been more substantial in South India, supporting the growth of the sugar industry.

Problems Faced by Sugar Industry in India:

- **Competition with Jaggery and Unrefined Sugar** - One of the primary challenges confronting the sugar industry is **intense competition from jaggery (gur) and unrefined sugar (Khandsari)**. Approximately **half of the sugarcane juice produced in India is utilized for making these alternatives**, leading to a reduced quantity of sugarcane available for sugar mills. This competition not only limits the supply of raw materials for sugar production but also affects the overall viability of sugar mills.
- **Low Yield per Hectare** - In comparison to conventional sugarcane-producing regions worldwide, such as **Hawaii, Cuba, and Mauritius**, India exhibits a notably **low per hectare yield of sugarcane**. For instance, while Hawaii boasts a remarkable yield of **173 tons per hectare**, Tamil Nadu achieves **107 tons per hectare**, and Uttar Pradesh lags significantly with only **62 tons per hectare**. This disparity highlights the inefficiencies present within the Indian sugarcane cultivation system, particularly when comparing yields between southern and northern states.
- **Sick Sugar Mills** - Another critical issue is the presence of **sick sugar mills** across the country. Many of these mills **face operational inefficiencies, financial difficulties, and lack of modernization**, which hinder their ability to compete effectively in the market. Addressing the problems of these sick mills is essential for revitalizing the overall health of the sugar industry.
- **Short Crushing Period** - The sugar industry is also plagued by a **short crushing period**, which limits the timeframe for sugarcane processing. Typically, the **crushing season lasts only a few months**, which constrains production capacity and **reduces the ability to maximize outputs** during the peak season.
- **Low Recovery Rate of Sugar** - The **low recovery rate of sugar** is another pressing concern. Currently, the average recovery rate in India is around **10%**, which is significantly lower than **14% in Australia**. This inefficiency translates into lower yields of refined sugar from the same quantity of sugarcane, adversely affecting the profitability of sugar mills.
- **Price Policy Challenges** - Finally, the **government's price policy is often perceived as unfavourable to sugar mill owners**. The regulatory framework surrounding sugar pricing **does not adequately support the interests of producers, leading to financial strain within the industry**. This lack of favourable pricing mechanisms undermines the overall growth potential of the sugar sector.

Jute Industry:

- The **jute industry** is a vital **agro-based sector** that primarily relies on the cultivation of jute as its raw material. As a **weight-losing industry**, the localization of jute production is concentrated in areas where raw jute is cultivated. The first jute mill in India was established in **Rishra**, located on the banks of the **Hugli River**, in **1855**. Before the partition of India in 1947, the country held a **monopoly over the global jute industry**. However, the partition negatively impacted this sector, as most of the jute mills (102 out of 112) remained in India while a significant portion of the jute-producing regions became part of **East Pakistan**, now known as **Bangladesh**.
- India is recognized as the **largest producer of raw jute and finished jute goods** globally. Furthermore, the country holds the **second position** in the world in the **export of finished jute products**. This prominent role underscores the importance of the jute industry to both the Indian economy and the global market.
- Currently, the jute industry is predominantly located in **West Bengal**, where the majority of jute mills are situated in the **Hugli industrial area** along both banks of the Hugli River. Key centers of jute production in this area include **Rishra, Bansbaria, Naihati, Chandan Nagar, Titagarh, Barrackpore, Howrah, Kolkata, Budge Budge, Manikpur**, and **Birlapur**.



Reasons for Concentration in West Bengal:

- **Raw Material Availability** - Approximately **76% of raw jute in India** is produced in this region, ensuring a consistent supply for the mills.
- **Fresh Water Supply** - The **Hugli delta** provides an abundant supply of fresh water, which is essential for extracting jute fibers.
- **Water Transportation** - The proximity to waterways facilitates efficient transport of both raw materials and finished products.
- **Energy Supply** - Access to **thermal and hydroelectric power** from the **Damodar Valley Corporation (DVC)** supports industrial operations.
- **Export Facilities** - The presence of the **Kolkata Harbour** provides vital facilities for exporting finished jute goods.
- **Labour Availability** - There is an abundance of **cheap labor** in the region, which is advantageous for the industry.
- **Infrastructural Development** - Existing **infrastructure** in West Bengal further supports the operations of jute mills.

Expansion Efforts Beyond West Bengal:

- While the jute industry is primarily concentrated in West Bengal, there has been **limited expansion** into other regions of India. **New areas for jute cultivation** have been **identified** through **agro-climatic regionalization**. These areas include:
 - **Loktak Basin** in **Manipur**
 - **Tripura Basin**
 - The **Terai area of Uttar Pradesh**
 - Delta regions of the **Mahanadi** and **Godavari** rivers

Bihar:

- Adjacent to West Bengal, the districts of **Purnia** and **Katihar** in **Bihar** are traditional areas for jute cultivation. Mills have been established in **Purnia**, **Katihar**, **Darbhanga**, and **Samastipur**.

Uttar Pradesh:

- In response to market demand, three jute mills have been established in **Uttar Pradesh**—two in **Kanpur** and one in **Sahjanwa** (Gorakhpur). These mills primarily produce jute bags for packaging **sugar and other agricultural products**, sourcing their raw materials from the jute cultivated in the **Terai region**.

Andhra Pradesh:

- Jute mills have also been set up in **Andhra Pradesh**, specifically in the districts of **Vishakhapatnam**, **Guntur**, and **East Godavari**.
- Additionally, jute mills have emerged in **Guwahati** (Assam), **Dimapur** (Nagaland), **Agartala** (Tripura), and **Raigarh** (Chhattisgarh).

Problems Faced by Jute Industry in India:

- **Declining Demand for Jute Products** - The **jute industry** is currently grappling with a significant decline in the demand for **finished jute goods**, both in **international** and **domestic markets**. This downturn can be primarily attributed to the **increasing competition from artificial fibres**, which have gained popularity due to their lower cost and versatility. As consumers shift towards synthetic alternatives, traditional jute products are facing challenges in maintaining their market presence.
- **Outdated Infrastructure and Resistance to Modernization** - Most of the existing jute mills are characterized by their **old and outdated machinery**. This antiquated infrastructure hampers efficiency and productivity, making it difficult for these mills to compete effectively in a modern market. However, the **mill owners** have shown reluctance towards **modernization** efforts. This resistance stems from the perception that investing in upgrades is not financially viable, as the jute industry is viewed as a **less profitable sector** compared to other manufacturing industries. Consequently, the lack of modernization further entrenches the industry's challenges.
- **Decrease in Jute Cultivation Areas** - Another pressing issue is the **decrease in conventional jute cultivation areas**. This reduction is largely due to the **expansion of hybrid paddy cultivation**, which is perceived as a more lucrative option for farmers. The shift in agricultural practices not only diminishes the available land for jute production but also affects the overall supply chain of the jute industry. As a result, the combination of reduced cultivation areas and decreased demand contributes to the ongoing struggles faced by the jute sector.

Forest-Based Industries:

Paper Industry:

- The **paper industry** in India has a rich history, marking its **beginnings in the early 19th century**. The **first attempt** to produce paper using modern techniques was made in **1816 in Tanjavur, Tamil Nadu**, but it proved to be unsuccessful. The first successful paper mill, however, was established in **1879 in Lucknow**, followed by additional mills in **Titagarh, West Bengal**, in **1881**. This period is recognized as the beginning of the modern paper industry in India.
- The paper industry is classified as a **weight-losing industry**, meaning it **requires a significant amount of raw materials to produce a relatively small quantity of finished product**. Approximately **2.5 tons of raw materials** are needed to manufacture just **one ton of paper**. Consequently, the localization of this industry predominantly occurs in regions abundant in these raw materials.

Raw Materials Used in Paper Production:

- **Softwood** - Softwood sourced from **coniferous trees** in the **Himalayan region** constitutes about **7%** of the total raw materials utilized in the paper industry.
- **Bamboo** - Bamboo is the most crucial raw material, contributing a staggering **70%** of the inputs for paper manufacturing in India. **Karnataka** is the leading producer of bamboo, followed closely by **Assam**.
- **Sabai Grass** - This grass provides **15%** of the raw material. Notably, the fibers of sabai grass are known for producing the **best quality paper**. The primary producer of sabai grass is **Madhya Pradesh**, with significant contributions from **Maharashtra, Andhra Pradesh**, and **Odisha**.
- **Bagasse** - This is the residual part of sugarcane and contributes **7%** of the pulp required for the paper industry. Bagasse is utilized in the production of **industrial paper, hardboard paper, and packing paper**.
- **Rags and Wastepaper** - The pulp for handmade paper is created from waste paper and rags. India excels in the production of **handmade paper** and is a significant exporter of this product. Handmade paper is often used for making **university certificates**, with **Puducherry** hosting Asia's largest handmade paper mill.

Distribution of the Paper Industry in India:

- The paper industry in India has evolved into a significant sector, with **various states contributing to its growth**. This distribution is largely **influenced by the availability of raw materials**, such as **bamboo, eucalypt, bagasse, and sabai grass**, which are crucial for paper production.

West Bengal:

- **West Bengal** is recognized as a traditional hub for the paper industry, with several paper mills concentrated in key locations. Major industrial centers include **Titagarh, Raniganj, Naihati, Bansbaria, Howrah, Kolkata**, and **Dumdum**. The raw materials for these mills are predominantly sourced from **north-west Bengal, Assam, Meghalaya**, and **Bangladesh**, providing a steady supply of inputs for paper production.

Maharashtra

- In **Maharashtra**, the paper industry is notably prominent, with major centers located in **Ballarpur** (home to the largest paper mill in India), **Kalyan, Khopoli, Chinchbud (Pune), Sangli, Kampati**, and **Jalgaon**. **Ballarpur** and **Sangli** are especially important for **newsprint production**. The industry in this state relies on various raw materials, including **bamboo, eucalyptus, bagasse, rags, and imported pulp**.

Karnataka

- **Karnataka** also plays a vital role in the paper industry, with mills established in **Bhadravati**, **Mysore**, **Bengaluru**, **Ramnagar**, and **Dandoli**. The primary raw material used in these mills is the local **bamboo**, ensuring a sustainable supply for production.

Andhra Pradesh and Telangana

- In **Andhra Pradesh**, the primary locations for paper mills are **Rajahmundry** and **Tirupati**, where **bamboo** and **sabai grasses** are predominantly utilized as raw materials. Similarly, **Telangana** has established mills mainly in **Sirpur**, **Kagaz Nagar**, **Khammam**, and **Patancheru**, supporting the region's paper production.

Odisha

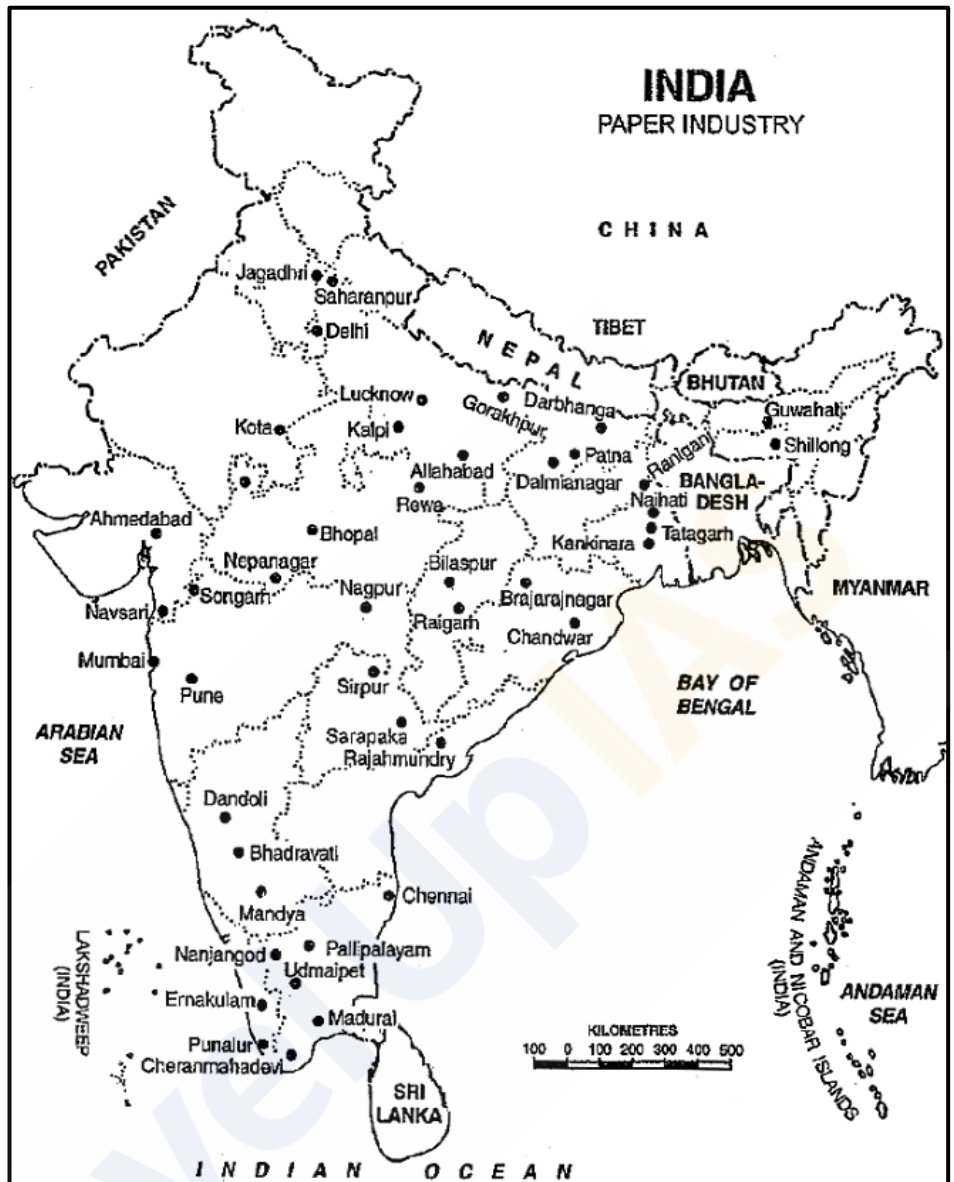
- **Odisha** contributes to the paper industry with major centers located in **Brajrajnagar**, **Raigora**, and **Balgopalpur**. These locations are crucial for the regional production of paper, taking advantage of local resources.

Madhya Pradesh

- **Madhya Pradesh** is known for its unique raw material base, primarily utilizing **sabai grass** for paper production. Major industrial centers in the state include **Bhopal**, **Indore**, **Ratlam**, and **Amlai**. Notably, **Nepanagar** is famous for **newsprint production**, while **Hoshangabad** is significant for manufacturing paper used in currency notes.

Other States

- **Haryana** - Major centers include **Yamuna Nagar**, **Faridabad**, and **Rohtak**.
- **Bihar** - Notable locations are **Dalmianagar** and **Samastipur**.
- **Kerala** - Paper mills are concentrated in **Calicut** and **Ernakulam**.
- **Tamil Nadu** - Key centers include **Salem** and **Tirunelveli**, with newspaper production occurring in **Pugalur**.
- **Assam** - Important locations are **Jorhat**, **Guwahati**, and **Kachhar**.
- **Uttar Pradesh** - Major centers are located in **Meerut**, **Saharanpur**, and **Modinagar**.



Match Industry:

- The **first match factory** in India was established in **1921** in **Ahmedabad**. Following this pioneering effort, additional match factories began to emerge in **1924 and 1925** in various locations including **Bareilly, Kolkata, Chennai, Ambarnath, and Dhubri**. This marked the beginning of a significant industry in the country, catering to the growing demand for matchsticks.

Raw Materials:

- Matchsticks are primarily manufactured from a special kind of **softwood**. The most suitable trees for producing this wood include **dhoop, markat, salai, semal, and sundari**. These varieties are known for their quality and suitability in match production. However, it is important to note that the **chemicals** necessary for the match-making process are typically **imported**, indicating a reliance on external sources for key components of the industry.

Regional Development:

- The match industry in India has seen significant development, particularly in the states of **West Bengal** and **Tamil Nadu**. In **West Bengal**, the industry is concentrated mainly in the districts of **24 Paragana** and **Kolkata**. Conversely, in **Tamil Nadu**, the majority of match factories are located in **Ramanathapuram district**, with additional factories in **Chingleput, Tirunelveli, and Chennai**.

Beyond these two primary states, match factories have also been established in other regions, including:

- **Maharashtra** - Thane, Pune, Chandrapur, and Mumbai
- **Gujarat** - Ahmedabad and Petlad
- **Uttar Pradesh** - Bareilly, Meerut, Allahabad, and Varanasi
- **Karnataka** - Shimoga
- **Telangana** - Hyderabad and Warangal
- **Assam** - Dhubri
- **Chhattisgarh** - Bilaspur
- **Rajasthan** - Kota

Silk Industry:

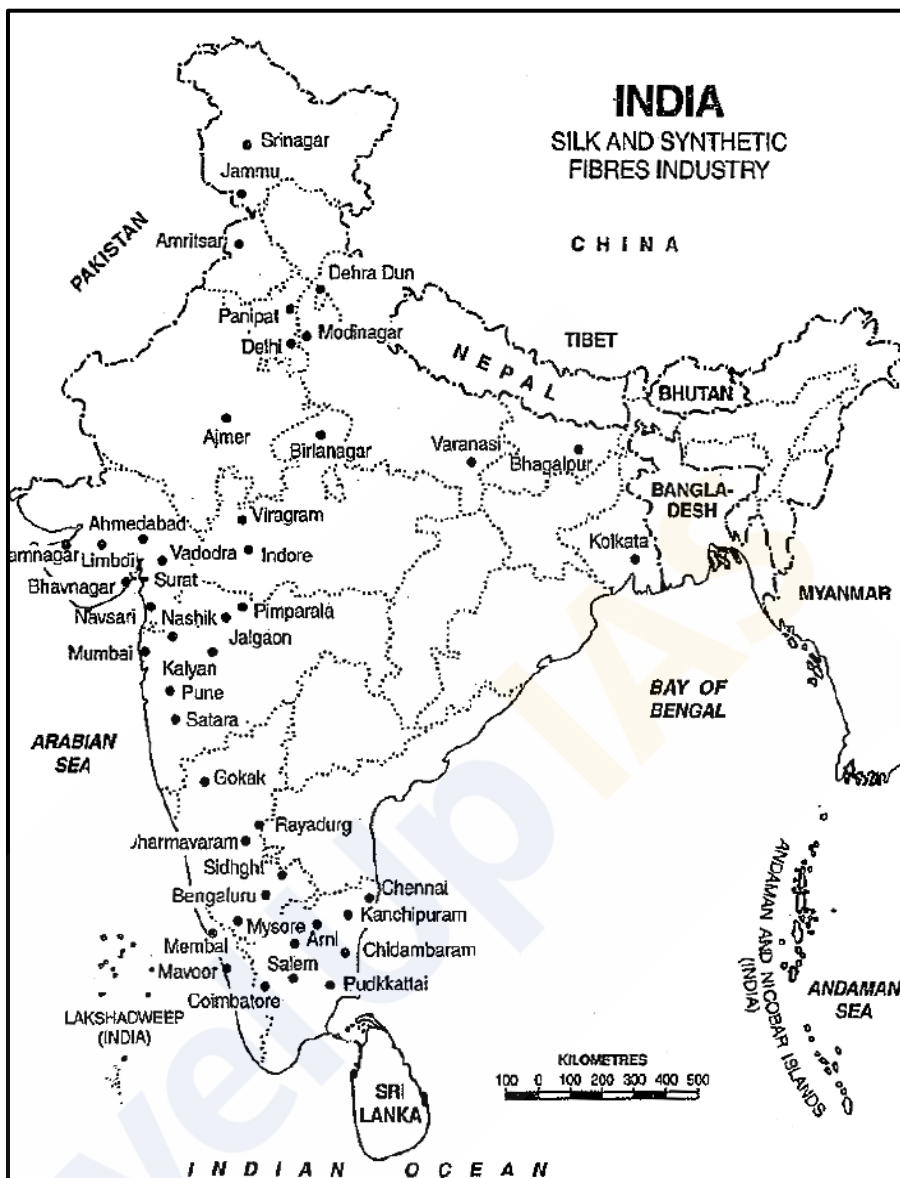
- The silk industry in India comprises two main stages: **sericulture** (the cultivation of silk-producing insects) and the **production of silk textiles** from silk fibers. Each stage plays a crucial role in the overall silk production process.

Sericulture:

- Sericulture is predominantly a **forest-based industry**, primarily conducted on **mulberry trees**. Other trees that support sericulture include **oak, mahua, castor, sal, plum, and kusum**. Notably, **Karnataka** leads the nation in silk production, contributing **45%** of the total output, followed by **Andhra Pradesh (29.5%)**, **West Bengal (9%)**, and **Tamil Nadu (8.2%)**.

Types of Silk Produced in India:

- **Mulberry Silk** - This type is derived from silkworms that feed exclusively on **mulberry leaves** and accounts for **72%** of India's silk production. Major production areas include **Karnataka** (specifically in **Bengaluru, Mysore, Kolar, and Tumkur districts**), along with regions in **West Bengal, Jammu & Kashmir, and Assam**.
- **Tussar Silk** - Produced from silkworms that thrive on leaves from various trees such as **dhak (Butea monosperma), kusum, mahua, sal, and arjun**. Tussar silk is characterized by its **light-yellow** colour and is considered inferior to mulberry silk. The city of **Bhagalpur in Bihar** is renowned for its handloom Tussar silk, with approximately **80%** of production occurring in **Jharkhand** and **11%** in **Chhattisgarh**.
- **Muga Silk** - Muga silk is produced by silkworms that feed on the leaves of **plum and arjun** trees. It is classified as **wild silk** and is notable for its **golden yellow** hue.
- **Eri Silk** - This silk is derived from silkworms that consume **castor leaves** and is recognized as a **lower quality silk**.



Silk Textile Industry:

The silk textile industry, while **indirectly connected to forest resources**, demonstrates three key tendencies in its localization:

- **Raw Material-Based Centers** - Locations such as **Mysore, Bengaluru, Coimbatore, and Srinagar** serve as centers for raw material production.
- **Demand-Based Centers** - Regions that attract demand, especially in religious contexts, include **Varanasi, Tirupati, Madurai, and Kanjivaram**.
- **Commercial or Collection Centers** - Industries in this category are established in towns like **Bhagalpur (Bihar), Erode, and Chennai (Tamil Nadu)**, as well as in **Mumbai**.

Karnataka:

- Karnataka stands out as the **largest producer of silk fibers** in India, playing a pivotal role in the silk industry.

Other Forest-Based Industries:

- The **wood sawing industry** flourishes primarily in **Kerala**, particularly in the **Calicut district**, which is famous for its sawmills.
- **Lac production** is another area where India excels, producing **60% of the world's lac** and emerging as the largest global exporter. Lac culture mainly occurs on **babool (acacia)** trees, with the **Chotanagpur area of Jharkhand** being a major contributor, accounting for about **50.83%** of India's total lac production. The **Lac Research Centre** in **Namkum (Ranchi)** is dedicated to the research and development of lac agriculture.
- The **beedi making industry**, largely established in tribal regions, sees **Madhya Pradesh** as the leading producer due to the abundance of **tendu leaves** in its forests.
- Additionally, **chemical industries** related to **sandalwood** in Karnataka and **eucalyptus** in Tamil Nadu have developed significantly.
- The **woodcraft industry** is prominent in **Karnataka**, **Himachal Pradesh**, and **Jammu and Kashmir**, while the **copra industry**, which processes coconut and its fibers, is managed by the **Coconut Board of India** in **Kerala**.

India's Industrial Policy:

India's **industrial policy** refers to the **government actions designed to influence the ownership, structure, and performance of the industrial sector**. It encompasses a range of mechanisms including **subsidies, financial support, regulation**, and other incentives or disincentives. The policy framework includes the **procedures, principles, rules, regulations**, and **tariff policies** that guide industrial development. It also outlines the government's **attitude toward foreign investment** and the **labour policy**, all aimed at fostering a conducive environment for industrial growth.

Objectives of Industrial Policy:

- To maintain a **sustained growth in productivity**.
- To enhance **gainful employment** opportunities.
- To achieve **optimal utilisation of human resources**.
- To attain **international competitiveness**.
- To transform India into a major player in the **global economic arena**.

Industrial Policies in India Since Independence:

Industrial Policy Resolution of 1948:

- The **Industrial Policy Resolution (IPR) of 1948** was the first comprehensive framework defining the role of the state in industrial development. It established a **mixed economic model** and classified industries into four categories:
 - **Strategic Industries (Public Sector)** - Included arms and ammunition, atomic energy, and rail transport, where the central government held a monopoly.
 - **Basic/Key Industries (Public-cum-Private Sector)** - Six industries such as coal, iron and steel, and aircraft manufacturing were identified as critical, primarily to be set up by the central government while allowing existing private enterprises to continue.
 - **Important Industries (Controlled Private Sector)** - Comprising 18 industries, this category was under private control but subject to government regulation.

- **Other Industries (Private and Cooperative Sector)** - All remaining industries were open to the private sector.
- The **Industries (Development and Regulation) Act of 1951** was enacted to implement this resolution.

Industrial Policy Statement of 1956:

- In **1956**, the government revised its industrial policy, known as the **Industrial Policy Statement of 1956**, which is often referred to as the "**Economic Constitution of India**". This policy focused on:
 - Expanding the **public sector**.
 - Promoting a **cooperative sector**.
 - Preventing the rise of **private monopolies**.
- The 1956 policy classified industries into three categories:
 - **Schedule A** - 17 industries exclusively under state control, including arms and ammunition and railways.
 - **Schedule B** - 12 industries open to both public and private sectors, but aimed for gradual state ownership.
 - **Schedule C** - All other industries not included in the first two schedules, primarily open to private investment.
- The **IPR of 1956** also highlighted the significance of **cottage and small-scale industries** for employment generation and the decentralization of economic power, calling for measures to maintain **industrial peace** and a fair distribution of production proceeds.
- **Criticism of the 1956 Policy** - The **IPR of 1956** faced criticism for severely limiting the expansion of the private sector due to stringent regulations and licensing requirements.

Industrial Policy Statement of 1977:

- In **1977**, the **Janata Government** introduced a new industrial policy focused on promoting **cottage and small industries** in rural areas. This policy categorized small-scale industries into three groups: cottage and household sector, tiny sector, and small-scale industries. It restricted the dominance of large business houses and emphasized **worker participation in management**.
- Despite its objectives, the 1977 policy was criticized for lacking effective measures to curb the dominance of large-scale units and for not envisioning significant socio-economic transformations.

Industrial Policy of 1980:

- The **Industrial Policy of 1980** aimed to enhance the efficiency of the public sector and reaffirmed the government's commitment to the **Monopolies and Restrictive Trade Practices (MRTP) Act** and the **Foreign Exchange Regulation Act (FERA)**.

New Industrial Policy During Economic Reforms of 1991:

- The **liberalized industrial policy** was introduced in **1991** amidst severe economic instability, with the aim of enhancing efficiency and accelerating economic growth. The key features included:
 - **De-reservation of Public Sector** - Reducing sectors exclusively reserved for public sector involvement, while retaining central control over crucial areas like arms and ammunition and railways.
 - **De-licensing** - Abolishing licensing requirements for most projects, with exceptions for industries related to security and the environment.
 - **Disinvestment of Public Sector** - Reducing government stakes in public enterprises to improve efficiency.

- **Liberalization of Foreign Investment** - Allowing foreign companies to have majority stakes in various sectors and enabling 100% Foreign Direct Investment (FDI) in several areas.
- **Foreign Technology Agreement** - Providing automatic approvals for technology-related agreements.
- **Amendments to MRTP Act** - Removing asset threshold limits for MRTP companies and replacing the MRTP Act with the **Competition Act 2002**.

Outcomes of New Industrial Policies:

- The 1991 policy made **'Licence, Permit and Quota Raj'** a thing of the past. It attempted to liberalise the economy by **removing bureaucratic hurdles** in industrial growth.
- **Limited role of Public sector** reduced the burden of the Government.
- The policy provided **easier entry of multinational companies**, privatisation, removal of asset limit on MRTP companies, liberal licensing.
 - All this resulted in increased competition, that led to lower prices in many goods such as electronics prices. This brought domestic as well as foreign investment in almost every sector opened to private sector.
- The policy was followed by **special efforts to increase exports**. Concepts like Export Oriented Units, Export Processing Zones, Agri-Export Zones, Special Economic Zones and lately National Investment and Manufacturing Zones emerged. All these have benefitted the export sector of the country.

Limitations of Industrial Policies in India:

- **Stagnation of Manufacturing Sector** - Industrial policies in India have failed to push manufacturing sector whose contribution to GDP is stagnated at about 16% since 1991.
- **Distortions in industrial pattern owing to selective inflow of investments** - In the current phase of investment following liberalisation, while substantial investments have been flowing into a few industries, there is concern over the slow pace of investments in many basic and strategic industries such as engineering, power, machine tools, etc.
- **Displacement of labour** - Restructuring and modernisation of industries as a sequel to the new industrial policy led to displacement of labour.
- **Absence of incentives for raising efficiency** - Focussing attention on internal liberalisation without adequate emphasis on trade policy reforms resulted in **'consumption-led growth' rather than 'investment' or 'export-led growth'**.
- **Vaguely defined industrial location policy** - The New Industrial Policy, while emphasised the detrimental effects of damage to the environment, failed to define a proper industrial location policy, which could ensure a pollution free development of industrial climate.

Way Forward:

- Industrial policies in India have taken a shift from predominantly **Socialistic pattern in 1956 to Capitalistic since 1991**.
- India now has a much-liberalised industrial policy regime focusing on increased foreign investment and lesser regulations.

- India ranked 77th on World Bank's Doing Business Report 2018. Reforms related to **insolvency resolution (Bankruptcy and Insolvency Act, 2017)** and the **Goods and Services Taxes (GST)** are impressive and will result in **long-term gains for the industrial sector**.
- Campaigns such as **Make in India** and **Start up India** have helped to enhance the business ecosystem in the country.
- However, electricity shortages and high prices, credit constraints, high unit labour costs due to labour regulations, political interference and other regulatory burdens continue to remain challenges for firm growth of the industrial sector in India.
- There is a **need for a new Industrial Policy** to boost the manufacturing sector in the country. Government in December 2018 also felt the need to introduce a new Industrial Policy that would be a road map for all business enterprises in the country.

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